

Product Specifications

(Structural Steel and Sheet Pile)

Type of Product	Classifications		Mechanical Properties								
			Yield Point N/mm ² (min.)		Tensile Strength N/mm ²	Yield Ratio % (max)	Elongation % (min)			Impact Thickness t > 12	
							Temp °C	Energy J (min)			
			Thickness			t ≥ 12			t ≤ 5	5 < t ≤ 15	t > 16
t ≤ 16	16 ≤ t ≤ 40										
Structural steel	JIS G 3101 : 2004 ^A	SS400	245	235	400-510	-	21	17	21	-	-
		SS490	285	275	490-610	-	19	15	19	-	-
		SS540	400	390	540 min	-	16	13	17	-	-
	JIS G 3106 : 2004	SM400 A	245	235	400-510	-	23	18	22	-	-
		SM400 B	245	235	400-510	-	23	18	22	0	27
		SM490 A	325	315	490-610	-	22	17	21	-	-
		SM490 B	325	315	490-610	-	22	17	21	0	27
		SM490 YA	365	355	490-610	-	19	15	19	-	-
		SM490 YB	365	355	490-610	-	19	15	19	0	27
		SM520 B	365	355	520-640	-	19	15	19	0	27
		SM520 C	365	355	520-640	-	19	15	19	0	47
	JIS G 3136 : 2005	SN400 A	235	235	400-510	-	17	17	21	-	-
		SN400 B	235-355 ^B	235	400-510	80 ^C	18	18	22	0	27
		SN490 B	325-445 ^B	235-355	490-610	80 ^C	17	17	21	0	27
	BS 4360 : 1986	43A	275	325-445	430-580	-		20		-	-
		43B	275	265	430-580	-		20		20	27G
		43C	275	265	430-580	-		20		0	27
		43D	275	265	430-580	-		20		-20	27
		50A	355	345	490-640	-		18		-	-
		50B	355	345	490-640	-		18		20	27G
		50C	355	345	490-640	-		18		0	27
		50D	355	345	490-640	-		18		-20	27
	BS EN 10025 : 1993/2004	S275JR	275	265	410-560	-		23		20	27G
		S275JO	275	265	410-560	-		23		0	27
		S275J2	275	265	410-560	-		23		-20	27
		S275J2G3E	275	265	410-560	-		23		-20	27
		S355JR	355	345	490 ^F -630	-		22		20	27G
S355JO		355	345	490 ^F -630	-		22		0	27	
S355J2		355	345	490 ^F -630	-		22		20	27	
S355J2G3E		355	345	470-630	-		22		20	27	
ASTM : 2003	A36	250		400-550	-		20		-	-	
	A 572 Gr.42	290		415 min	-		20		-	-	
	A 572 Gr.50	345		450 min ^D	-		18		-	-	
	A 992	345-450		450 min	85		18		-	-	
DIN 17100	St 33	185	175	290	-		16		-	-	
	St 37-2	235	225	340-470	-		24		-	-	
	St 44-2	275	265	410-540	-		20		-	-	
	St 50-2	295	285	470-610	-		18		-	-	
	St 52-3	355	345	490-630	-		20		-	-	
AS/NZS 3679.1 : 1996		t < 12	12 < t ≤ 40								
	250	260	250	410 min	-		22		-	-	
	250L0	260	250	410 min	-		22		-	-	
	350	360	340	480 min	-		20		-	-	
	350L0	360	340	480 min	-		20		-	-	
		t < 11	11 ≤ t ≤ 17	t < 17							
Sheet Pile	JIS A 5528 : 2000	SY295	295		490 min	-		17		-	-
		SY390	390		540 min	-		15		-	-

Remark

A: Bend test on material grades SS400, SS490, SS540.

B: For the H section, when the t₁ is 9 mm or less, the upper limit of the yield point or proof stress shall not be applied.

C: For the H section, when the t₁ is 9 mm or less, the upper limit of the yield ratio shall be 85%.

D: For grades 50 steel of thickness 20 mm and under, the tensile strength shall be a minimum of 485 N/mm².

E: Only in 1993 version.

F: For 2004 version, the specified tensile strength is 470-630 N/mm².

G: Verified only when specified at the time of the order.

Product Specifications

(Structural Steel and Sheet Pile)

Chemical Compositions (ladle analysis), %											Classifications	
C (max.)	Si (max.)	Mn (max.)	P (max.)	S (max.)	CE ^E (max.)	SW ^E (max.)	Nb (max.)	V (max.)	N (max.)			
-	-	-	0.050	0.050	-	-	-	-	-	-	SS400	JIS G 3101 : 2004
-	-	-	0.050	0.050	-	-	-	-	-	-	SS490	
0.30	-	1.60	0.040	0.040	-	-	-	-	-	-	SS540	
0.23	-	250xC min.	0.035	0.035	-	-	-	-	-	-	SM400 A	JIS G 3106 : 2004
0.20	0.35	0.60-1.40	0.035	0.035	-	-	-	-	-	-	SM400 B	
0.20	0.55	1.60	0.035	0.035	-	-	-	-	-	-	SM490 A	
0.18	0.55	1.60	0.035	0.035	-	-	-	-	-	-	SM490 B	
0.20	0.55	1.60	0.035	0.035	-	-	-	-	-	-	SM490 YA	
0.20	0.55	1.60	0.035	0.035	-	-	-	-	-	-	SM490 YB	
0.20	0.55	1.60	0.035	0.035	-	-	-	-	-	-	SM520 B	
0.20	0.55	1.60	0.035	0.035	-	-	-	-	-	-	SM520 C	
0.24	-	-	0.050	0.050	-	-	-	-	-	-	SN400 A	JIS G 3136 : 2005
0.20	0.35	0.60-1.40	0.030	0.015	0.36	0.36	-	-	-	-	SN400 B	
0.18	0.55	1.60	0.030	0.015	0.44	0.44	-	-	-	-	SN490 B	
0.25	0.50	1.60	0.050	0.050	-	-	-	-	-	-	43A	BS 4360 : 1986
0.21	0.50	1.50	0.050	0.050	-	-	-	-	-	-	43B	
0.18	0.50	1.50	0.050	0.050	-	-	-	-	-	-	43C	
0.18	0.50	1.50	0.050	0.050	-	-	-	-	-	-	43D	
0.23	0.50	1.60	0.050	0.050	-	-	0.003-0.100	0.003-0.100	-	-	50A	
0.20 ^A	0.50	1.50 ^A	0.050	0.050	-	-	0.003-0.100	0.003-0.100	-	-	50B	
0.20 ^A	0.50	1.50 ^A	0.050	0.050	-	-	0.003-0.100	0.003-0.100	-	-	50C	
0.18 ^A	0.50	1.50 ^A	0.040	0.040	-	-	0.003-0.100	0.003-0.100	-	-	50D	
0.21	-	1.50	0.035	0.035	-	-	-	-	0.009 ^C	-	S275JR	BS EN 10025 : 1993/2004 ^H
0.18	-	1.50	0.030	0.030	-	-	-	-	0.009 ^C	-	S275JO	
0.18	-	1.50	0.025	0.025	-	-	-	-	-	-	S275J2	
0.18	-	1.50	0.035	0.035	-	-	-	-	-	-	S275J2G3 ^G	
0.24	0.55	1.60	0.035	0.035	-	-	-	-	0.009 ^C	-	S355JR	
0.20	0.55	1.60	0.030	0.030	-	-	-	-	0.009 ^C	-	S355JO	
0.20	0.55	1.60	0.035	0.035	-	-	-	-	-	-	S355J2	
0.20	0.55	1.60	0.035	0.035	-	-	-	-	-	-	S355J2G3 ^G	
0.26	0.40	-	0.040	0.050	-	-	-	-	-	-	A36	ASTM : 2003
0.21	0.40	1.35 ^B	0.040	0.050	-	-	-	0.010-0.150	-	-	A 572 Gr.42	
0.23	0.40	1.35 ^B	0.040	0.050	-	-	-	0.010-0.150	-	-	A 572 Gr.50	
0.23	0.10-0.40	0.50-1.50	0.035	0.045	0.45	-	-	0.11	0.012	-	A 992 ^D	
-	-	-	-	-	-	-	0.05	-	-	-	St-33	DIN 17100
0.17	-	-	0.050	0.050	-	-	-	-	0.009 ^C	-	St 37-2	
0.21	-	-	0.050	0.050	-	-	-	-	0.009 ^C	-	St 44-2	
-	-	-	0.050	0.050	-	-	-	-	0.009 ^C	-	St 50-2	
0.2	-	-	0.040	0.040	-	-	-	-	-	-	St 52-3	
0.25	0.40	-	0.040	0.040	0.43	0.43	-	-	-	-	250	AS/NZS 3679.1 : 1996
0.20	0.40	1.50	0.040	0.040	0.42	0.42	-	-	-	-	250L0	
0.22	0.50	1.60	0.040	0.040	0.45	0.45	-	-	-	-	350	
0.22	0.50	1.60	0.040	0.040	0.45	0.45	-	-	-	-	350L0	
0.22	0.50	1.60	0.040	0.040	0.44	0.44	-	-	-	-	300	
0.22	0.50	1.60	0.040	0.040	0.44	0.44	-	-	-	-	300L0	
-	-	-	0.040	0.040	-	-	-	-	-	-	SY295	JIS A 5528 : 2000
-	-	-	0.040	0.040	-	-	-	-	-	-	SY390	

Remark

A: It is permissible to vary the carbon and manganese contents (ladle analysis) for grades 50B, 50C, and 50D on the basis of an increase of 0.06% manganese content for each decrease of 0.01% carbon, or vice versa up to maximum manganese content of 1.6% and maximum carbon content of 0.22% for grades 50B and 50C and 0.20% for grade 50D

B: A maximum of manganese of 1.50% is permissible, with an associated reduction of the carbon maximum of 0.03%.

C: Nitrogen control excluded.

D: Additional chemical composition contro grade A992 : Ni (max.) = 0.45, Co (max.) = 0.05, Cr (max.) = 0.35, Cu (max.) = 0.60, Mo (max.) = 0.15, Co +V (max.) = 0.15, and Mn/S (min.) = 20, Mn/S (min.) = 20.

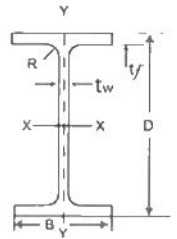
E: Carbon Equivalent : (JIS Standard) CE = C + Mn/6 + Si/24 + Ni/40 + Cr/5 + Mo/4 + V/14 (AS, ASTM Standard) CE = C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15.

F: Chmical Composition on sensitivity of welding crack = C + Si/30 + Mn/20 + Cu/20 + Ni/60 + Cr/20 + Mo/15 + V/10 + 5B.

G: Only in 1993 version.

H: For 2004 version, Cu (max.) = 0.55%

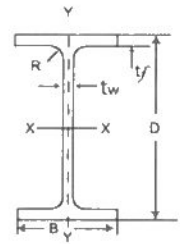
Universal Beams And Columns



IMPERIAL UNITS

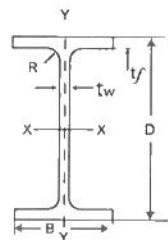
Section Number and Nominal Size	Unit weight M		Section area A		Section depth D		Flange				Web Thickness tw		Corner radius r	
							Width B		Thickness T					
mm (in.)	lb/ft	kg/m	in. ²	cm ²	in.	mm	in.	mm	in.	mm	in	mm	in	mm
W4 102 x 102 (4 x 4)	16.3 13.8 13	24.0 21.0 19.35	4.79 4.05 3.83	30.9 26.1 24.71	4.21 4.02 4.16	107.0 102.0 105.7	3.937 4.016 4.060	100.0 102.0 103.1	0.472 0.370 0.345	12.0 9.4 8.76	0.311 0.315 0.280	7.9 8.0 7.11	0.236 0.236 0.25	6.0 6.0 6.35
W5 127 x 76 (5 x 3)	9	13.00	2.60	16.80	5.00	127.0	3.000	76.2	0.299	7.60	0.165	4.20	0.30	6.35
W5 127 x 127 (5 x 5)	19 16	28.28 23.81	5.56 4.71	35.87 30.39	5.15 5.01	130.8 127.3	5.030 5.000	127.8 127.0	0.430 0.360	10.90 9.14	0.270 0.240	6.86 6.10	0.30 0.30	7.62 7.62
W6 152 x 76 (6 x 3)	12 9	18.0 14.0	3.35 2.76	21.6 17.8	6.10 5.91	155.0 150.0	2.953 2.953	75.0 75.0	0.335 0.276	8.5 7.0	0.236 0.197	6.0 5.0	0.315 0.315	8.0 8.0
W6 152 x 89 (6 x 3½)	11	16.00	3.18	20.50	6.00	152.4	3.500	88.9	0.303	7.70	0.181	4.60	0.25	6.35
W6 152 x 102 (6 x 4)	16 12 9	23.81 17.86 13.39	4.74 3.55 2.68	30.58 22.90 17.29	6.28 6.03 5.90	159.5 153.2 149.9	4.030 4.000 3.940	102.4 101.6 100.1	0.405 0.280 0.215	10.20 7.11 5.46	0.260 0.230 0.170	6.60 5.84 4.32	0.25 0.25 0.25	6.35 6.35 6.35
W6 152 x 152 (6 x 6)	25 20 15.7 15	37.20 30.00 23.00 22.50	7.34 5.87 4.62 4.43	47.35 37.87 29.80 28.58	6.38 6.20 6.00 5.99	162.1 157.5 152.4 152.1	6.080 6.020 6.000 5.990	154.4 152.9 152.4 152.1	0.455 0.365 0.269 0.260	11.50 9.27 6.80 6.60	0.320 0.260 0.240 0.230	8.13 6.60 6.10 5.84	0.25 0.25 0.25 0.25	6.35 6.35 6.35 6.35
W7 178 x 89 (7 x 3½)	15 12 11	22.2 18.1 16.1	4.37 3.56 3.16	28.2 23.0 20.4	7.05 6.89 6.81	179.0 175.0 173.0	3.543 3.543 3.543	90.0 90.0 90.0	0.394 0.315 0.276	10.0 8.0 7.0	0.236 0.197 0.177	6.0 5.0 4.5	0.35 0.35 0.35	8.9 8.9 8.9
W7 178 x 102 (7x4)	13	19.00	3.75	24.20	7.00	177.8	4.000	101.6	0.311	7.90	0.185	4.70	0.30	7.62
W8 203 x 102 (8 x 4)	15.5 15 13 10	23.00 22.32 19.35 14.88	4.50 4.44 3.84 2.96	29.00 28.65 24.77 19.10	8.00 8.11 7.99 7.89	203.2 206.0 202.9 200.4	4.000 4.015 4.000 3.940	101.6 102.0 101.6 100.1	0.368 0.315 0.255 0.205	9.30 8.00 6.48 5.21	0.205 0.245 0.230 0.170	5.20 6.22 5.84 4.32	0.30 0.30 0.30 0.30	7.62 7.62 7.62 7.62
W8 203 x 133 (8 x 5¼)	21 20 16 17 15 14	31.30 30.00 26.79 25.30 22.3 21.0	6.16 5.89 5.26 5.01 4.45 4.20	39.74 38.00 33.94 32.30 28.7 27.1	8.28 8.14 8.14 8.00 7.95 7.99	210.5 206.8 206.8 203.2 202.0 203.0	5.270 5.268 5.250 5.250 5.236 5.236	133.9 133.8 133.3 133.4 133.0 133.0	0.400 0.378 0.330 0.308 0.276 0.252	10.10 9.60 8.38 7.80 7.0 6.4	0.250 0.248 0.230 0.230 0.197 0.197	6.35 6.30 5.84 5.84 5.0 5.0	0.30 0.30 0.30 0.30 0.31 0.31	7.62 7.62 7.62 7.62 8.0 8.0
W8 203 x 165 (8 x 6½)	28 24	41.67 35.72	8.25 7.08	53.23 45.68	8.06 7.93	204.7 201.4	6.535 6.495	166.0 165.0	0.465 0.400	11.80 10.10	0.285 0.245	7.24 6.22	0.40 0.40	10.2 10.2
W8 203 x 203 (8 x 8)	67 58 48 40 35 31	100.0 86.31 71.43 60.00 52.09 46.13	19.7 17.1 14.1 11.7 10.3 9.12	127.1 110.3 90.97 75.48 66.45 58.84	9.00 8.75 8.50 8.25 8.12 8.00	228.6 222.2 215.9 209.5 206.2 203.2	8.280 8.220 8.110 8.070 8.020 7.995	210.3 208.8 206.0 205.0 203.7 203.1	0.935 0.810 0.685 0.560 0.495 0.435	23.75 20.57 17.40 14.20 12.50 11.00	0.570 0.510 0.400 0.360 0.310 0.285	14.48 12.95 10.16 9.14 7.87 7.24	0.40 0.40 0.40 0.40 0.40 0.40	10.2 10.2 10.2 10.2 10.2 10.2
W10 254 x 102 (10 x 4)	19 17 15 12	28.28 25.30 22.32 17.86	5.62 4.99 4.41 3.54	36.26 32.19 28.45 22.84	10.24 10.11 9.99 9.87	260.1 256.8 253.7 250.7	4.020 4.010 4.000 3.960	102.1 101.9 101.6 100.6	0.395 0.330 0.270 0.210	10.00 8.38 6.86 5.33	0.250 0.240 0.230 0.190	6.35 6.10 5.84 4.83	0.30 0.30 0.30 0.30	7.62 7.62 7.62 7.62
W10 254 x 146 (10 x 5¼)	30 29 26 25 22 21 16	44.80 43.00 38.69 37.20 32.74 31.25 24.0	8.85 8.54 7.61 7.36 6.49 6.20 4.82	57.10 55.10 49.10 47.50 41.87 40.00 31.1	10.47 10.22 10.33 10.08 10.17 9.90 9.96	265.9 259.6 262.4 256.0 258.3 251.5 253.0	5.819 5.799 5.770 5.762 5.750 5.750 5.709	147.8 147.3 146.6 146.4 146.0 146.1 145.0	0.510 0.500 0.440 0.430 0.360 0.340 0.252	12.90 12.70 11.10 10.90 9.14 8.60 6.4	0.300 0.289 0.260 0.252 0.240 0.240 0.197	7.62 7.30 6.60 6.40 6.10 6.10 5.0	0.30 0.30 0.30 0.30 0.30 0.30 0.30	7.62 7.62 7.62 7.62 7.62 7.62 7.6
W10 254 x 203 (10 x 8)	57 45 42 39 33	84.8 66.97 62.5 58.04 49.11	16.74 13.3 12.29 11.5 9.71	108 85.81 79.3 74.19 62.64	8.15 10.10 9.89 9.92 9.73	207.0 256.5 249.4 252.0 247.1	8.315 8.020 8.150 7.985 7.960	211.3 203.7 207.0 202.8 202.2	0.650 0.620 0.480 0.530 0.435	16.5 15.70 12.2 13.40 11.00	0.650 0.350 0.480 0.315 0.290	16.5 8.89 12.2 8.00 7.37	0.50 0.50 0.50 0.50 0.50	12.7 12.7 12.7 12.7 12.7
W10 254 x 254 (10 x 10)	112 100 89	167.0 149.0 132.0	32.9 29.4 26.2	212.3 189.7 168.9	11.36 11.11 10.88	288.5 282.2 276.4	10.415 10.340 10.275	264.5 262.6 261.0	1.250 1.120 0.998	31.75 28.45 25.30	0.755 0.680 0.615	19.18 17.27 15.60	0.50 0.50 0.50	12.7 12.7 12.7

Universal Beams And Columns (Continued)



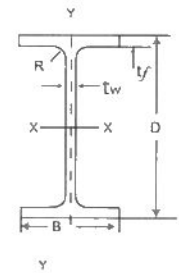
Section Number and Nominal size	Moment of inertia				Radius of gyration				Modulus of section			
	I_x		I_y		i_x		i_y		Z_x		Z_y	
mm ⁴ /(in ⁴)	in. ⁴	cm ⁴	in. ⁴	cm ⁴	in.	cm	in.	cm	in. ³	cm ³	in. ³	cm ³
W4 102 x 102 (4 x 4)	14.14 11.0 11.3	587 456 470	4.80 4.01 3.86	200 167 161	1.72 1.65 1.72	4.36 4.18 4.37	1.00 1.00 1.00	2.55 2.53 2.54	6.71 5.46 5.46	110 89.4 89.5	2.45 2.00 1.90	40.1 32.7 31.1
W5 127 x 76 (5 x 3)	11.5	477	1.30	56.0	2.10	5.33	0.72	1.83	4.60	75.0	0.90	15.0
W5 127 x 127 (5 x 5)	26.3 21.4	1,090 891	9.13 7.51	380 313	2.17 2.13	5.51 5.41	1.2 1.26	3.25 3.20	10.2 8.55	167 140	3.63 3.00	59.50 49.2
W6 152 x 76 (6 x 3)	20.2 16.0	841 666	1.45 1.19	60.2 49.5	2.46 2.41	6.24 6.11	0.66 0.66	1.67 1.67	6.65 5.42	109 88.8	0.98 0.81	16.0 13.2
W6 152 x 89 (6 x 3½)	20.1	838	2.20	90.0	2.52	6.40	0.83	2.10	6.70	110	1.20	20.0
W6 152 x 102 (6 x 4)	32.1 22.1 16.4	1,340 920 683	4.43 2.99 2.20	184 124 91.6	2.60 2.49 2.47	6.60 6.32 6.27	0.966 0.918 0.905	2.45 2.33 2.30	10.2 7.31 5.56	167 120 91.1	2.20 1.50 1.11	36.1 24.6 18.2
W6 152 x 152 (6 x 6)	53.4 41.4 30.3 29.1	2,220 1,720 1,263 1,210	17.1 13.3 9.70 9.32	712 554 403 388	2.70 2.66 2.56 2.56	6.86 6.76 6.51 6.50	1.52 1.50 1.45 1.45	3.86 3.81 3.68 3.68	16.7 13.4 10.1 9.72	274 220 165.7 159	5.61 4.41 3.20 3.11	91.9 72.3 52.95 51.0
W7 178 x 89 (7 x 3½)	36.7 29.1 25.5	1,529 1,213 1,060	2.93 2.34 2.05	122 97.6 85.4	2.90 2.86 2.83	7.36 7.26 7.20	0.82 0.81 0.80	2.08 2.06 2.04	10.43 8.48 7.51	171 139 123	1.65 1.32 1.16	27.1 21.7 19.0
W7 178 x 102 (7 x 4)	32.6	1,357	3.30	138	2.95	7.49	0.940	2.39	9.30	153	1.60	27.0
W8 203 x 102 (8 x 4)	50.2 48.0 39.6 30.8	2,091 2,000 1,650 1,280	3.90 3.41 2.73 2.09	163 142 114 87.0	3.34 3.29 3.21 3.22	8.49 8.36 8.16 8.18	0.930 0.876 0.843 0.841	2.37 2.22 2.14 2.14	12.6 11.8 9.91 7.81	206 193 162 128	2.00 1.70 1.37 1.06	32.0 27.9 22.4 17.4
W8 203 x 133 (8 x 5¼)	75.3 69.2 61.9 56.4 50.6 47.6	3,130 2,981 2,580 2,348 2,105 1,980	9.77 8.50 7.97 6.72 6.61 6.03	407 354 332 280 275 251	3.49 3.43 3.43 3.36 3.22 3.20	8.86 8.71 8.71 8.53 8.56 8.55	1.26 1.20 1.23 1.16 1.22 1.20	3.20 3.05 3.12 2.95 3.09 3.05	18.2 17.0 15.2 14.1 12.7 11.9	298 279 249 231 208 195	3.71 3.20 3.04 2.60 2.52 2.31	60.8 52.4 49.8 42.6 41.3 37.8
W8 203 x 165 (8 x 6½)	98.0 82.8	4,080 3,450	21.7 18.3	903 762	3.45 3.42	8.76 8.69	1.62 1.61	4.11 4.09	24.3 20.9	398 342	6.63 5.63	109 92.3
W8 203 x 203 (8 x 8)	272 228 184 146 127 110	11,300 9,480 7,660 6,080 5,290 4,580	88.6 75.1 60.9 49.1 42.6 37.1	3,690 3,130 2,530 2,040 1,770 1,540	3.72 3.65 3.61 3.53 3.51 3.47	9.45 9.27 9.17 8.97 8.92 8.81	2.12 2.10 2.08 2.04 2.03 2.02	5.38 5.33 5.28 5.18 5.16 5.13	60.4 52.0 43.3 35.5 31.2 27.5	990 852 710 582 511 451	21.4 18.3 15.0 12.2 10.6 9.27	351 300 246 200 174 152
W10 254 x 102 (10 x 4)	96.3 81.9 68.9 53.8	4,010 3,410 2,870 2,240	4.29 3.56 2.89 2.18	176 148 120 90.7	4.14 4.05 3.95 3.90	10.5 10.3 10.0 9.91	0.874 0.845 0.810 0.785	2.22 2.15 2.06 1.99	18.8 16.2 13.8 10.9	308 265 226 179	2.14 1.78 1.45 1.10	35.1 29.2 23.8 18.0
W10 254 x 146 (10 x 5¼)	170 157 144 133 118 106 83.5	7,080 6,548 5,990 5,545 4,910 4,425 3,477	16.8 15.2 14.1 12.7 11.4 9.7 7.83	699 633 587 529 474 404 326	4.38 4.29 4.35 4.26 4.27 4.14 4.17	11.1 10.9 11.0 10.8 10.8 10.5 10.6	1.38 1.34 1.36 1.31 1.33 1.25 1.27	3.51 3.40 3.45 3.33 3.38 3.18 3.23	32.5 30.8 27.9 26.4 23.2 21.5 16.78	533 505 457 433 380 352 275	5.76 5.20 4.89 4.40 3.97 3.40 2.74	94.4 85.2 80.1 72.1 65.1 55.7 44.9
W10 254 x 203 (10 x 8)	286.5 248 202.5 209 170	11,930 10,300 8,430 8,700 7,080	62.61 53.4 43.41 45.0 36.6	2,606 2,220 1,807 1,870 1,520	4.13 4.32 4.06 4.27 4.19	10.5 11.0 10.3 10.8 10.6	1.93 2.01 1.88 1.98 1.94	4.91 5.11 4.77 5.03 4.93	56.44 49.1 41.25 42.1 35.0	905 805 676 690 574	15.07 13.3 10.68 11.3 9.20	247 218 175 185 151
W10 254 x 254 (10 x 10)	716 625 542	29,800 26,000 22,580	236 207 181	9,820 8,620 7,518	4.66 4.61 4.55	11.8 11.7 11.6	2.68 2.65 2.63	6.81 6.73 6.68	126 112 99.7	2,060 1,840 1,634	45.3 40.0 35.2	742 655 577

Universal Beams And Columns



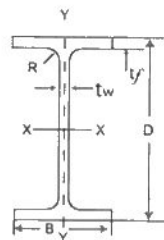
Section Number and Nominal Size	Unit weight M		Section area A		Section depth D		Flange				Web Thickness tw		Corner radius r	
							Width B		Thickness T					
mm (in.)	lb/ft	kg/m	in ²	cm ²	in.	mm	in.	mm	in.	mm	in.	mm	in	mm
W10 254 x 254 (10 x 10)	88	131.0	25.9	167.1	10.84	275.3	10.265	260.7	0.990	25.15	0.605	15.37	0.50	12.7
	77	115.0	226	145.8	10.60	269.2	10.190	258.8	0.870	22.10	0.530	13.46	0.50	12.7
	72	107.0	21.2	136.6	10.50	266.7	10.170	258.3	0.808	20.50	0.510	13.00	0.50	12.7
	68	101.2	20.0	129.0	10.40	264.2	10.130	257.3	0.770	19.56	0.470	11.94	0.50	12.7
	60	89.29	17.6	113.5	10.22	259.6	10.080	256.0	0.680	17.27	0.420	10.67	0.50	12.7
	54	80.36	15.8	101.9	10.09	256.3	10.030	254.8	0.615	15.60	0.370	9.40	0.50	12.7
49	73.00	14.4	92.90	9.98	253.5	10.000	254.0	0.560	14.20	0.340	8.64	0.50	12.7	
W12 305 x 102 (12 x 4)	22	33.00	6.48	41.81	12.31	312.7	4.030	102.4	0.425	10.80	0.260	6.60	0.30	7.62
	19	28.28	5.57	35.94	12.16	308.9	4.005	101.7	0.350	8.89	0.235	5.97	0.30	7.62
	16.5	25.00	4.87	31.40	12.00	304.8	4.000	101.6	0.269	6.80	0.230	5.80	0.30	7.62
	16	23.81	4.71	30.39	11.99	304.5	3.990	101.3	0.265	6.73	0.220	5.59	0.30	7.62
14	20.83	4.16	26.84	11.91	302.5	3.970	100.8	0.225	5.71	0.200	5.08	0.30	7.62	
W12 305 x 127 (12 x 5)	32	48.00	9.42	60.8	12.22	310.4	4.930	125.2	0.551	14.00	0.350	8.90	0.30	7.62
	28	42.00	8.25	53.2	12.07	306.6	4.893	124.3	0.476	12.10	0.313	8.00	0.30	7.62
	25	37.00	7.36	47.5	11.96	303.8	4.864	123.5	0.421	10.70	0.284	7.20	0.30	7.62
W12 305 x 165 (12 x 6½)	36	54.00	10.6	68.40	12.24	310.9	6.565	166.8	0.540	13.70	0.305	7.70	0.30	7.62
	35	52.09	10.3	66.45	12.50	317.5	6.560	166.6	0.520	13.20	0.300	7.62	0.30	7.62
	31	46.00	9.13	58.90	12.09	307.1	6.525	165.7	0.465	11.80	0.265	6.70	0.30	7.62
	30	44.64	8.79	56.71	12.34	313.4	6.520	165.6	0.440	11.10	0.260	6.60	0.30	7.62
	27	40.00	7.98	51.50	11.96	303.8	6.500	165.1	0.400	10.20	0.240	6.10	0.30	7.62
	26	38.70	7.65	49.35	12.22	310.4	6.490	164.8	0.380	9.65	0.230	5.84	0.30	7.62
	21	31.0	6.11	39.4	12.05	306.0	6.460	164.0	0.290	7.4	0.196	5.0	0.30	7.62
W12 305 x 203 (12 x 8)	50	74.41	14.7	94.84	12.19	309.6	8.080	205.2	0.640	16.20	0.370	9.40	0.60	15.2
	45	66.97	13.2	85.16	12.06	306.3	8.045	204.3	0.575	14.60	0.335	8.51	0.60	15.2
	40	59.53	11.8	76.13	11.94	303.3	8.005	203.3	0.515	13.00	0.295	7.49	0.60	15.2
W12 305 x 254 (12 x 10)	58	86.31	17.0	109.7	12.19	309.6	10.010	254.3	0.640	16.20	0.360	9.14	0.60	15.2
	53	78.87	15.6	100.6	12.06	306.3	9.995	253.9	0.575	14.60	0.345	8.76	0.60	15.2
W12 305 x 305 (12 x 12)	336	500.0	98.8	637.4	16.82	427.2	13.385	340.0	2.955	75.06	1.775	45.08	0.60	15.2
	305	453.9	89.6	578.1	16.32	414.5	13.235	336.2	2.705	68.71	1.625	41.27	0.60	15.2
	278	413.7	81.9	528.4	15.85	402.6	13.140	333.8	2.470	62.74	1.530	38.86	0.60	15.2
	242	375.0	74.0	477.4	15.41	391.4	13.005	330.3	2.250	57.15	1.395	35.43	0.60	15.2
	230	342.3	67.7	436.8	15.05	382.3	12.895	327.5	2.070	52.58	1.285	32.64	0.60	15.2
	210	312.5	61.8	398.7	14.71	373.6	12.790	324.9	1.900	48.26	1.180	29.97	0.60	15.2
	190	283.0	55.8	360.0	14.38	365.3	12.670	321.8	1.735	44.07	1.060	26.92	0.60	15.2
	170	253.0	50.0	322.6	14.03	356.4	12.570	319.3	1.560	39.62	0.960	24.38	0.60	15.2
	161	240.0	47.4	305.6	13.88	352.6	12.515	317.9	1.486	37.70	0.905	23.00	0.60	15.2
	152	226.2	44.7	288.4	13.71	348.2	12.480	317.0	1.400	35.56	0.870	22.10	0.60	15.2
	136	202.4	39.9	257.4	13.41	340.6	12.400	315.0	1.250	31.75	0.790	20.07	0.60	15.2
	133	198.0	39.1	252.3	13.38	339.9	12.365	314.1	1.236	31.40	0.755	19.20	0.60	15.2
	120	179.0	34.3	227.7	13.12	333.2	12.320	312.9	1.100	28.07	0.710	18.03	0.60	15.2
	106	158.0	31.2	201.3	12.89	327.4	12.220	310.4	0.990	24.14	0.610	15.48	0.60	15.2
	96	143.0	28.2	182.0	12.71	322.8	12.160	308.9	0.900	22.86	0.550	13.97	0.60	15.2
	92	137.0	27.1	174.6	12.62	320.4	12.155	308.7	0.856	21.70	0.545	13.80	0.60	15.2
	87	129.5	25.6	165.2	12.53	318.3	12.125	308.0	0.810	20.57	0.515	13.08	0.60	15.2
	79	118.0	23.2	149.7	12.38	314.5	12.080	306.8	0.735	18.67	0.470	11.94	0.60	15.2
	72	107.1	21.2	136.1	12.25	311.1	12.040	305.8	0.670	17.02	0.430	10.92	0.60	15.2
65	97.0	19.1	123.2	12.12	307.8	12.000	304.8	0.605	15.30	0.390	9.91	0.60	15.2	
W14	26	39.0	7.69	49.61	13.91	353.3	5.025	127.6	0.420	10.60	0.255	6.48	0.40	10.2
356 x 127 (14 x 5)	22	33.0	6.49	41.87	13.74	349.0	5.000	127.0	0.335	8.51	0.230	5.84	0.40	10.2
W14 356 x 171 (14 x 6½)	45	67.0	13.2	85.40	14.33	364.0	6.820	173.2	0.618	15.70	0.357	9.10	0.40	10.2
	38	57.0	11.2	72.26	14.10	358.1	6.770	172.0	0.515	13.00	0.310	7.87	0.40	10.2
	34	51.0	10.0	64.52	13.98	355.1	6.745	171.3	0.455	11.50	0.285	7.24	0.40	10.2
	30	45.0	8.85	57.10	13.84	351.5	6.730	170.9	0.385	9.78	0.270	6.86	0.40	19.2

Universal Beams And Columns (Continued)



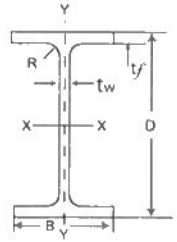
Section Number and Nominal size	Moment of inertia				Radius of gyration				Modulus of section			
	I_x		I_y		i_x		i_y		Z_x		Z_y	
mm/(in.)	in. ⁴	cm ⁴	in. ⁴	cm ⁴	in.	cm	in.	cm	in. ³	cm ³	in. ³	cm ³
W10	534	22,200	179	7,450	4.54	11.5	2.63	6.68	98.5	1,610	34.8	570
254 x 254	455	18,900	154	6,410	4.49	11.4	2.60	6.60	85.9	1,410	30.1	493
(10 x 10)	421	17,510	142	5,903	4.46	11.3	2.59	6.58	80.1	1,312	27.9	457
	394	16,400	134	5,580	4.44	11.3	2.58	6.55	75.7	1,240	26.4	433
	341	14,200	116	4,830	4.39	11.2	2.57	6.53	66.7	1,090	23.0	377
	303	12,600	103	4,290	4.37	11.1	2.56	6.50	60.0	983	20.6	338
	272	11,300	93.4	3,890	4.35	11.0	2.54	6.45	54.6	895	18.7	306
W12	156	6,490	4.66	194	4.91	12.5	0.848	2.15	25.4	416	2.31	37.9
305 x 102	130	5,410	3.76	156	4.82	12.2	0.822	2.09	21.3	349	1.88	30.8
(12 x 4)	105.4	4,387	2.90	120	4.65	11.8	0.770	1.96	17.6	288	1.40	23.6
	103	4,290	2.82	117	4.67	11.9	0.773	1.96	17.1	280	1.41	23.1
	88.6	3,690	2.36	98.2	4.62	11.7	0.753	1.91	14.9	244	1.19	19.5
W12	228.3	9,504	11.1	460	4.92	12.5	1.08	2.75	37.4	612	4.50	73.5
305 x 126	195.6	8,143	9.3	388	4.87	12.4	1.06	2.70	32.4	531	3.80	62.5
(12 x 5)	172.0	7,162	8.1	337	4.83	12.3	1.05	2.67	28.8	472	3.30	54.6
W12	285	11,900	24.5	1,020	5.25	13.3	1.54	3.91	45.6	747	7.47	122
305 x 165	280.8	11,690	23.7	986.6	5.15	13.1	1.50	3.81	45.9	752	7.20	118
(12 x 6½)	238.4	9,924	19.6	824	5.11	12.9	1.47	3.73	39.4	646	6.10	100
	238	9,910	20.3	845	5.21	13.2	1.52	3.86	38.6	633	6.24	102
	204.1	8,496	16.6	691	5.06	12.8	1.44	3.66	34.1	559	5.10	83.6
	204	8,490	17.3	720	5.17	13.1	1.51	3.84	33.4	547	5.34	87.5
	157.5	6,554	13.1	544	5.08	12.9	1.46	3.72	26.1	428	4.05	66.4
W12	394	16,400	56.3	2,340	5.18	13.2	1.96	4.98	64.7	1,060	13.9	228
305 x 203	350	14,600	50.0	2,080	5.15	13.1	1.94	4.93	58.1	952	12.4	203
(12 x 8)	310	12,900	44.1	1,840	5.13	13.0	1.94	4.93	51.9	850	11.0	180
W12	475	19,800	107	4,450	5.28	13.4	2.51	6.38	78.0	1,280	21.4	351
305 x 254 (12 x 10)	425	17,700	95.8	3,990	5.23	13.3	2.48	6.30	70.6	1,160	19.2	315
W12	4,060	169,000	1,190	49,500	6.41	16.3	3.47	8.81	483	7,910	177	2,900
305 x 305	3,550	148,000	1,050	43,700	6.29	16.0	3.42	8.69	435	7,130	159	2,610
(12 x 12)	3,110	129,000	937	39,000	6.16	15.6	3.38	8.59	393	6,440	143	2,340
	2,720	113,000	828	34,500	6.06	15.4	3.34	8.48	353	5,780	127	2,080
	2,420	101,000	742	30,900	5.97	15.2	3.31	8.41	321	5,260	115	1,880
	2,140	89,100	664	27,600	5.89	15.0	3.28	8.33	292	4,780	104	1,700
	1,890	78,700	589	24,500	5.82	14.8	3.25	8.25	263	4,310	93.0	1,520
	1,542	64,180	486	20,240	5.70	14.5	3.20	8.13	222	3,641	77.1	1,273
	1,540	68,700	517	21,500	5.74	14.6	3.22	8.18	235	3,850	82.3	1,350
	1,430	59,500	454	18,900	5.66	14.4	3.19	8.10	209	3,420	72.8	1,190
	1,240	51,600	398	16,600	5.58	14.2	3.16	8.03	186	3,050	64.2	1,050
	1,221	50,840	390	16,230	5.59	14.2	3.16	8.03	183	2,991	63.1	1,034
	1,070	44,500	345	14,400	5.51	14.0	3.13	7.95	163	2,670	56.0	918
	933	38,800	301	12,500	5.47	13.9	3.11	7.90	145	2,380	49.3	808
	833	34,700	270	11,200	5.44	13.8	3.09	7.85	131	2,150	44.4	728
	789	32,840	256	10,670	5.40	13.7	3.08	7.82	125	2,048	42.2	691
	740	30,800	241	10,000	5.38	13.7	3.07	7.80	118	1,930	39.7	651
	662	27,600	216	8,990	5.34	13.6	3.05	7.75	107	1,750	35.8	587
	597	24,800	195	8,120	5.31	13.5	3.04	7.72	97.4	1,600	32.4	531
	533	22,200	174	7,240	5.28	13.4	3.02	7.67	87.9	1,440	29.1	477
W14	245	10,200	8.91	371	5.65	14.4	1.08	2.74	35.3	578	3.55	58.2
356 x 127 (14 x 5)	199	8,280	7.00	291	5.54	14.1	1.04	2.64	29.0	475	2.80	45.9
W14	469	19,522	32.7	1,362	5.95	15.1	1.57	3.99	54.4	1,073	9.60	157
356 x 171 (14 x 6½)	385	16,000	26.7	1,110	5.88	14.9	1.55	3.94	54.6	895	7.88	129
	340	14,200	23.3	970	5.83	14.8	1.53	3.89	48.6	796	6.91	113
	291	12,100	19.6	816	5.73	14.6	1.49	3.78	42.0	688	5.82	95.4

Universal Beams And Columns



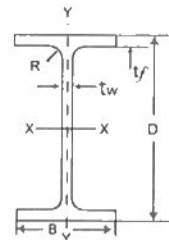
Section Number and Nominal Size	Unit weight M		Section area A		Section depth D		Flange				Web Thickness tw		Corner radius r	
							Width B		Thickness T					
mm (in.)	lb/ft	kg/m	in ²	cm ²	in.	mm	in.	mm	in	mm	in.	mm	in.	mm
W14 356 x 203 (14 x 8)	53 48 43	78.87 71.43 63.99	15.6 14.1 12.6	100.6 90.97 81.29	13.92 13.79 13.66	353.6 350.3 347.0	8.060 8.030 7.995	204.7 204.0 203.1	0.660 0.595 0.530	16.70 15.10 13.40	0.370 0.340 0.305	9.40 8.64 7.75	0.60 0.60 0.60	15.2 15.2 15.2
W14 356 x 254 (14 x 10)	82 74 68 61	122.0 110.1 101.2 90.78	24.1 21.8 20.0 17.9	155.5 140.6 129.0 115.5	14.31 14.17 14.04 13.89	363.5 359.9 356.6 352.8	10.130 10.070 10.035 9.995	257.3 255.8 254.9 253.9	0.855 0.785 0.723 0.645	21.72 19.94 18.36 16.30	0.510 0.450 0.415 0.375	12.95 11.43 10.54 9.52	0.60 0.60 0.60 0.60	15.2 15.2 15.2 15.2
W14 356 x 368 (14 x 14½)	136 132 120 119 109 103 99 90 87	202.0 196.4 179.0 177.0 162.2 153.0 147.3 134.0 129.0	40.0 38.8 35.3 35.0 32.0 30.3 29.1 26.5 25.6	257.9 250.3 227.7 225.7 206.5 195.2 187.7 171.0 164.9	14.75 14.66 14.48 14.50 14.32 14.25 14.16 14.02 14.00	374.7 372.4 367.8 368.3 363.7 362.0 359.7 356.1 355.6	14.740 14.725 14.670 14.550 14.605 14.575 14.565 14.520 14.500	374.4 374.0 372.6 372.1 371.0 370.2 270.0 368.8 368.3	1.063 1.030 0.940 0.938 0.860 0.813 0.780 0.710 0.688	27.00 26.16 23.88 23.80 21.84 20.70 19.81 18.03 17.50	0.660 0.645 0.590 0.570 0.525 0.495 0.485 0.440 0.420	16.80 16.38 14.99 14.50 13.33 12.60 12.32 11.18 10.70	0.60 0.60 0.60 0.60 0.60 0.60 0.60 0.60 0.60	15.2 15.2 15.2 15.2 15.2 15.2 15.2 15.2 15.2
W14 356 x 406 (14 x 16)	730 665 605 550 500 455 426 398 370 342 314 311 283 264 257 233 228 211 193 176 159 158 145 118	1,086.0 989.6 900.3 818.5 744.1 677.1 634.0 592.3 551.0 509.0 467.0 463.0 421.1 393.0 382.5 347.0 340.0 314.0 287.2 262.0 237.0 235.0 216.0 175.6	215 196 178 162 147 134 125 117 109 101 92.3 91.4 83.3 77.6 75.6 68.5 67.1 62.0 56.8 51.8 46.7 46.5 42.7 34.7	1,387 1,265 1,148 1,045 948.4 864.5 806.4 754.8 703.2 651.6 595.5 589.7 537.4 500.9 487.7 441.9 432.7 400.0 366.4 334.2 301.3 299.8 275.5 224	22.42 21.64 20.92 20.24 19.60 19.02 18.67 18.29 17.92 17.54 17.19 17.12 16.74 16.50 16.38 16.04 16.00 15.72 15.48 15.22 14.98 15.00 14.78 14.17	569.5 549.7 531.4 514.1 497.8 483.1 474.2 464.6 455.2 445.5 436.6 434.8 425.2 419.1 416.1 407.4 406.4 399.3 393.2 386.6 380.5 381.0 375.4 360.0	17.890 17.650 17.415 17.200 17.010 16.835 16.695 16.590 16.475 16.360 16.235 16.230 16.110 16.025 15.995 15.890 15.865 15.800 15.710 15.650 15.565 15.550 15.500 15.590	454.4 448.3 442.3 436.9 432.1 427.6 424.1 421.4 418.5 415.5 412.4 412.2 409.2 407.0 406.3 403.6 403.0 401.3 399.0 397.5 395.4 395.0 393.7 396.0	4.910 4.520 4.160 3.820 3.500 3.210 3.035 2.845 2.660 2.470 2.283 2.260 2.070 1.938 1.890 1.720 1.688 1.560 1.440 1.310 1.190 1.188 1.090 0.787	124.70 114.80 105.60 97.03 88.90 81.53 77.09 72.26 67.56 62.74 58.00 57.40 52.58 49.20 48.01 43.69 42.90 39.62 36.58 33.27 30.23 30.20 27.69 20.00	3.070 2.830 2.595 2.380 2.190 2.015 1.875 1.770 1.655 1.540 1.415 1.410 1.290 1.205 1.175 1.070 1.045 0.980 0.890 0.830 0.745 0.730 0.680 0.780	77.98 71.88 65.91 60.45 55.63 51.18 47.62 44.96 42.04 39.12 35.90 35.81 32.77 30.60 29.84 27.18 26.50 24.89 22.61 21.08 18.92 18.50 17.27 20.00	0.60 0.59	15.2 15.0
W16 406 x 140 (16 x 5½)	31 26	46.13 39.00	9.12 7.58	58.84 49.55	15.88 15.69	403.4 398.5	5.525 5.500	140.3 139.7	0.440 0.345	11.10 8.76	0.275 0.250	6.98 6.35	0.40 0.40	10.2 10.2
W16 406 x 178 (16 x 7)	57 50 45 40 36	85.0 74.41 67.0 60.0 54.0	16.8 14.7 13.3 11.8 10.6	108.4 94.84 85.81 76.13 68.39	16.43 16.26 16.13 16.01 15.86	417.3 413.0 409.7 406.7 402.8	7.120 7.070 7.035 6.995 6.985	180.8 179.6 178.7 177.7 177.4	0.715 0.630 0.565 0.505 0.430	18.16 16.00 14.30 12.80 10.90	0.430 0.380 0.345 0.305 0.295	10.92 9.65 8.76 7.75 7.49	0.40 0.40 0.40 0.40 0.40	10.2 10.2 10.2 10.2 10.2
W16 406 x 260 (16 x 10¼)	100 89 77 67	148.8 132.4 114.6 99.71	29.4 26.2 22.6 19.7	189.7 169.0 145.8 127.1	16.97 16.75 16.52 16.33	431.0 425.4 419.6 414.8	10.425 10.365 10.295 10.235	264.8 263.3 261.5 260.0	0.985 0.875 0.760 0.665	25.02 22.22 19.30 16.89	0.585 0.525 0.455 0.395	14.86 13.33 11.56 10.03	0.40 0.40 0.40 0.40	10.2 10.2 10.2 10.2

Universal Beams And Columns (Continued)



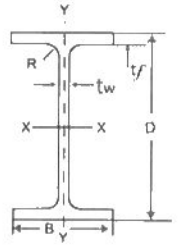
Section Number and Nominal size	Moment of inertia				Radius of gyration				Modulus of section			
	I_x		I_y		r_x		r_y		Z_x		Z_y	
mm/(in)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in	cm
W14 356 x 203 (14 x 8)	542 485 428	22,600 20,200 17,800	57.7 51.4 45.2	2,400 2,140 1,880	5.89 5.85 5.82	15.0 14.9 14.8	1.92 1.91 1.89	4.88 4.85 4.80	77.8 70.3 62.7	1,270 1,150 1,030	14.3 12.8 11.3	234 210 185
W14 356 x 254 (14 x 10)	882 796 725 640	36,700 33,100 30,200 26,600	148 134 122 107	6,160 5,580 5,080 4,450	6.05 6.04 6.01 5.98	15.4 15.3 15.3 15.2	2.48 2.48 2.47 2.45	6.30 6.30 6.27 6.22	123 112 103 92.2	2,020 1,840 1,690 1,510	29.3 26.6 24.3 21.5	480 436 398 352
W14 356 x 368 (14 x 14½)	1,593 1,530 1,380 1,373 1,240 1,166 1,110 999 966	63,310 63,700 57,400 57,160 51,600 48,530 46,200 41,600 40,250	568 548 495 492 447 420 402 362 350	23,630 22,800 20,600 20,470 18,600 17,470 16,700 15,100 14,560	6.31 6.28 6.24 6.26 6.22 6.21 6.17 6.14 6.15	16.0 16.0 15.8 15.9 15.8 15.8 15.7 15.6 15.6	3.77 3.76 3.74 3.75 3.73 3.72 3.71 3.70 3.70	9.58 9.55 9.50 9.52 9.47 9.45 9.42 9.40 9.40	216 209 190 189 173 164 157 143 138	3,540 3,420 3,110 3,104 2,830 2,681 2,570 2,340 2,263	77 74.5 67.5 67.1 61.2 57.6 55.2 49.9 48.2	1,262 1,220 1,110 1,099 1,000 944 905 818 790
W14 356 x 406 14 x 16	14,300 12,400 10,800 9,430 8,210 7,190 6,600 6,000 5,440 4,900 4,399 4,330 3,840 3,526 3,400 3,010 2,942 2,660 2,400 2,140 1,900 1,900 1,710 1,244	595,000 516,000 450,000 392,000 342,000 299,000 275,000 250,000 226,000 204,000 183,100 180,000 160,000 146,800 142,000 125,000 122,500 111,000 99,900 89,100 79,100 79,100 71,200 51,770	4,720 4,170 3,680 3,250 2,880 2,560 2,360 2,170 1,990 1,810 1,631 1,610 1,440 1,331 1,290 1,150 1,124 1,030 931 838 748 745 677 498	196,000 174,000 153,000 135,000 120,000 107,000 98,200 90,300 82,800 75,300 67,910 67,000 59,900 55,420 53,700 47,900 46,820 42,900 38,800 34,900 31,100 31,010 28,200 20,730	8.17 7.98 7.80 7.63 7.48 7.33 7.26 7.16 7.07 6.98 6.90 6.88 6.79 6.74 6.71 6.63 6.62 6.55 6.50 6.43 6.38 6.40 6.33 5.98	20.8 20.3 19.8 19.4 19.0 18.6 18.4 18.2 18.0 17.7 17.5 17.5 17.2 17.1 17.0 16.8 16.8 16.6 16.5 16.3 16.2 16.2 16.1 15.2	4.69 4.62 4.55 4.49 4.43 4.38 4.34 4.31 4.27 4.24 4.20 4.20 4.17 4.14 4.13 4.10 4.10 4.07 4.05 4.02 4.00 4.00 3.98 3.78	11.9 11.7 11.6 11.4 11.3 11.1 11.0 10.9 10.8 10.8 10.7 10.7 10.6 10.5 10.5 10.4 10.4 10.3 10.3 10.2 10.2 10.1 9.61	1,280 1,150 1,040 931 838 756 706 656 607 559 512 506 459 427 415 375 368 338 310 281 254 253 232 175	21,000 18,800 17,000 15,300 13,700 12,400 11,600 10,700 9,950 9,160 8,390 8,290 7,520 7,005 6,800 6,150 6,028 5,540 5,080 4,600 4,160 4,153 3,800 2,876	627 472 423 378 339 304 283 262 241 221 201 199 179 166 161 145 141 130 119 107 96.2 95.8 87.3 63.9	8,640 7,730 6,930 6,190 5,560 4,980 4,640 4,290 3,950 3,620 3,294 3,260 2,930 2,772 2,640 2,380 2,324 2,130 1,950 1,750 1,580 1,570 1,430 1,047
W16 406 x 140 (16 x 5½)	375 301	15,600 12,500	12.4 9.59	516 399	6.41 6.25	16.3 15.9	1.17 1.12	2.97 2.84	47.2 38.4	773 629	4.49 3.49	73.6 57.2
W16 406 x 178 (16 x 7)	758 659 586 518 448	31,600 27,400 24,400 21,600 18,600	43.1 37.2 32.8 28.9 24.5	1,790 1,550 1,370 1,200 1,020	6.72 6.68 6.65 6.63 6.51	17.1 17.0 16.9 16.8 16.5	1.60 1.59 1.57 1.57 1.52	4.06 4.04 3.99 3.99 3.86	92.2 81.0 72.7 64.7 56.5	1,510 1,330 1,190 1,060 926	12.1 10.5 9.34 8.25 7.00	198 172 153 135 115
W16 406 x 260 (16 x 10¼)	1,490 1,300 1,100 954	62,000 54,100 46,200 39,700	186 163 138 119	7,740 6,780 5,740 4,956	7.10 7.05 7.00 6.96	18.0 17.9 17.8 17.7	2.51 2.49 2.47 2.46	6.38 6.32 6.27 6.25	175 155 134 117	2,870 2,540 2,200 1,920	35.7 31.4 26.9 23.2	585 515 441 380

Universal Beams And Columns



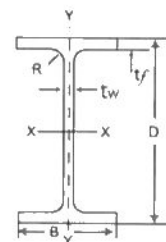
Section Number and Nominal Size	Unit weight M		Section area A		Section depth D		Flange				Web Thickness tw		Corner radius r	
							Width B		Thickness T					
mm (in)	lb/ft	kg/m	in ²	cm ²	in	mm	in	mm	in	mm	in	mm	in	mm
W18 457 x 152 (18 x 6)	55	82.00	16.2	104.5	18.31	465.1	6.042	153.5	0.744	18.90	0.420	10.7	0.40	10.2
	50	74.00	14.7	95.00	18.16	461.3	6.011	152.7	0.669	17.00	0.389	9.90	0.40	10.2
	46	68.46	13.5	87.10	18.06	458.7	6.060	153.9	0.605	15.30	0.360	9.14	0.40	10.2
	45	67.00	13.2	85.40	18.00	457.2	5.982	151.9	0.589	15.00	0.360	9.10	0.40	10.2
	40	60.00	11.8	76.13	17.90	454.7	6.015	152.8	0.525	13.30	0.315	8.00	0.40	10.2
	35	52.09	10.3	66.45	17.70	449.6	6.000	152.4	0.425	10.80	0.300	7.62	0.40	10.2
W18 457 x 191 (18 x 7½)	71	106.0	20.8	134.2	18.47	469.1	7.635	193.9	0.810	20.57	0.495	12.57	0.40	10.2
	66	98.00	19.4	125.3	18.40	467.4	7.592	192.8	0.770	19.60	0.450	11.40	0.40	10.2
	65	97.00	19.1	123.2	18.35	466.1	7.590	192.8	0.750	19.05	0.450	11.43	0.40	10.2
	60	89.29	17.6	113.5	18.24	463.3	7.555	191.9	0.695	17.65	0.415	10.54	0.40	10.2
	55	82.00	16.2	104.5	18.11	460.0	7.530	191.3	0.630	16.00	0.390	9.91	0.40	10.2
	50	74.41	14.7	94.84	17.99	456.9	7.495	190.4	0.570	14.40	0.355	9.02	0.40	10.2
	45	67.00	13.2	85.40	17.86	453.6	7.476	189.9	0.500	12.70	0.334	8.50	0.40	10.2
	41	61.00	11.8	76.00	17.72	450.0	7.441	189.0	0.425	10.80	0.319	8.10	0.40	10.2
W18 457 x 279 (18 x 11)	175	260.0	51.3	331.0	20.04	509.0	11.378	289.0	1.591	40.40	0.890	22.60	0.40	10.2
	119	177.1	35.1	226.5	18.97	481.8	11.265	286.1	1.060	26.92	0.655	16.64	0.40	10.2
	106	157.7	31.1	200.6	18.73	475.7	11.200	284.5	0.940	23.88	0.590	14.99	0.40	10.2
	97	144.4	28.5	183.9	18.59	472.2	11.145	283.1	0.870	22.10	0.535	13.59	0.40	10.2
	86	128.0	25.3	163.2	18.39	467.1	11.090	281.7	0.770	19.56	0.480	12.19	0.40	10.2
	76	113.1	22.3	143.9	18.21	462.5	11.035	280.3	0.680	17.27	0.425	10.80	0.40	10.2
W21 533 x 165 (21 x 6½)	57	85.00	16.7	107.7	21.06	534.9	6.555	166.5	0.650	16.51	0.405	10.29	0.50	12.7
	50	74.41	14.7	94.84	20.83	529.1	6.530	165.9	0.583	13.50	0.380	9.65	0.50	12.7
	44	66.00	13.0	83.87	20.66	524.8	6.500	165.1	0.450	11.40	0.350	8.89	0.50	12.7
W21 533 x 210 (21 x 8¼)	93	138.4	27.3	176.1	21.62	549.1	8.420	213.9	0.930	23.62	0.580	14.73	0.50	12.7
	83	123.5	24.3	156.8	21.43	544.3	8.355	212.2	0.835	21.21	0.515	13.08	0.50	12.7
	82	122.0	24.1	155.8	21.44	544.6	8.342	211.9	0.840	21.30	0.502	12.80	0.50	12.7
	73	109.0	21.5	138.7	21.24	539.5	8.295	210.7	0.740	18.80	0.455	11.56	0.50	12.7
	68	101.2	20.0	129.0	21.13	536.7	8.270	210.1	0.685	17.40	0.430	10.92	0.50	12.7
	62	92.27	18.3	118.1	20.99	533.1	8.240	209.3	0.615	15.62	0.400	10.16	0.50	12.7
	55	82.00	16.2	104.4	20.80	528.3	8.216	208.7	0.520	13.20	0.376	9.60	0.50	12.7
	48	71.4	14.1	91.0	20.60	523.2	8.142	206.8	0.429	10.90	0.350	8.90	0.50	12.7
W21 533 x 311 (21 x 12¼)	147	218.8	43.2	278.7	22.06	560.3	12.510	317.8	1.150	29.21	0.720	18.29	0.50	12.7
	132	196.4	38.8	250.3	21.83	554.5	12.440	316.0	1.030	26.29	0.650	16.51	0.50	12.7
	122	181.6	35.9	231.6	21.68	550.7	12.390	314.7	0.960	24.38	0.600	15.24	0.50	12.7
	111	165.2	32.7	211.0	21.51	546.4	12.340	313.4	0.875	22.22	0.550	13.97	0.50	12.7
	101	150.3	29.8	192.3	21.36	542.5	12.290	312.2	0.800	20.32	0.500	12.70	0.50	12.7
W24 610 x 178 (24 x 7)	62	92.27	18.2	117.4	23.74	603.0	7.040	178.8	0.590	14.99	0.430	10.92	0.50	12.7
	61	91.00	17.9	116.5	23.72	602.5	7.023	178.4	0.591	15.00	0.419	10.60	0.50	12.7
	55	82.00	16.2	104.5	23.57	598.7	7.005	177.9	0.505	12.83	0.395	10.03	0.50	12.7
W24 610 x 229 (24 x 9)	146	217	43.1	278	25.32	643.1	9.217	234.1	1.382	35.1	0.772	19.6	0.50	12.7
	128	191	37.5	242	25.00	635.0	9.114	231.5	1.216	30.9	0.669	17.0	0.50	12.7
	114	171	33.8	218	24.76	628.9	9.051	229.9	1.098	27.9	0.610	15.5	0.50	12.7
	103	153	30.2	195	24.53	623.1	9.000	228.6	0.980	24.9	0.551	14.0	0.50	12.7
	94	140.0	27.7	178.7	24.31	598.7	9.065	230.2	0.875	22.22	0.515	13.08	0.50	12.7
	84	125.0	24.7	159.4	24.10	617.5	9.020	229.1	0.770	19.56	0.470	11.94	0.50	12.7
	76	113.1	22.4	144.5	23.92	612.1	8.990	228.3	0.680	17.27	0.440	11.18	0.50	12.7
	68	101.2	20.1	129.7	23.73	607.6	8.965	227.7	0.585	14.86	0.415	10.54	0.50	12.7
W24 610 x 305 (24 x 12)	160	238	47.0	303.8	24.92	636.0	12.264	311.5	1.235	31.4	0.732	18.6	0.50	17.8
	120	179	35.3	227.9	24.31	617.5	12.088	307.0	0.930	23.6	0.556	14.1	0.50	17.8
	100	149	29.4	190.1	24.00	609.6	12.000	304.8	0.775	19.7	0.468	11.9	0.70	17.8
W24 610 x 324 (24 x 12¼)	192	285.7	56.3	363	25.86	656.9	12.949	328.9	1.461	37.1	0.811	20.6	0.50	12.7
	176	261.9	51.6	333	25.24	641.1	12.890	327.4	1.339	34.0	0.752	19.1	0.50	12.7
	162	241.1	47.7	307.7	25.00	635.0	12.955	329.1	1.220	30.99	0.705	17.91	0.50	12.7
	146	217.3	43.0	277.4	24.74	628.4	12.900	327.7	1.090	27.69	0.650	16.51	0.50	12.7
	131	194.9	38.5	248.4	24.48	621.8	12.855	326.5	0.960	24.38	0.605	15.37	0.50	12.7
	117	174.1	34.4	221.9	24.26	616.2	12.800	325.1	0.850	21.59	0.550	13.97	0.50	12.7
	104	154.8	30.6	197.4	24.06	611.1	12.750	323.8	0.750	19.05	0.500	12.70	0.50	12.7

Universal Beams And Columns (Continued)



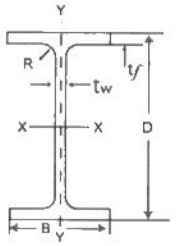
Section Number and Nominal size	Moment of inertia				Radius of gyration				Modulus of section			
	I_x		I_y		r_x		r_y		Z_x		Z_y	
mm (in)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
W18 457 x 152 (18 x 6)	870 779 712 686 612 510	36,215 32,435 29,600 28,577 25,500 21,200	27.5 24.3 22.5 21.1 19.1 15.3	1,143 1,012 937 878 795 637	7.33 7.28 7.25 7.20 7.21 7.04	18.6 18.5 18.4 18.3 18.3 17.9	1.30 1.28 1.29 1.26 1.27 1.22	3.31 3.26 3.28 3.21 3.23 3.10	95.0 85.8 78.8 76.3 68.4 57.6	1,557 1,406 1,290 1,250 1,120 944	31.5 8.10 7.43 7.00 6.35 5.12	149 133 122 116 104 83.9
W18 457 x 191 (18 x 7½)	1,170 1,098 1,070 984 890 800 704	48,700 45,717 44,500 41,000 37,000 33,300 29,330 25380	60.3 56.3 54.8 50.1 44.9 40.1 31.9	2,510 2,343 2,280 2,090 1,870 1,670 1,328 1218	7.50 7.52 7.49 7.47 7.41 7.38 7.30 7.00	19.0 19.1 19.0 19.0 18.8 18.7 18.5 18.2	1.70 1.70 1.69 1.68 1.67 1.65 1.55 1.55	4.32 4.33 4.29 4.27 4.24 4.19 3.94 3.99	127 119 117 108 98.3 88.9 78.9	2,080 1,956 1,920 1,770 1,610 1,460 1,293 1128	15.8 14.8 14.4 13.3 11.9 10.7 8.5	259 243 236 128 195 175 139 129
W18 457 x 279 (18 x 11)	3,452 2,190 1,910 1,750 1,530 1,330	143,700 91,200 79,500 72,800 63,700 55,400	392 253 220 201 175 152	16,300 10,500 9,170 8,370 7,280 6,330	8.19 7.90 7.84 7.82 7.77 7.73	20.8 20.1 19.9 19.9 19.7 19.6	2.76 2.69 2.66 2.65 2.63 2.61	7.01 6.83 6.76 6.73 6.68 6.63	345 231 204 188 166 146	5,646 3,790 3,340 3,080 2,720 2,390	68.3 44.9 39.4 36.1 31.6 27.6	1,128 736 646 592 518 452
W21 533 x 165 (21 x 6½)	1,170 984 843	48,700 41,000 35,100	30.6 24.9 20.7	1,270 1,040 862	8.36 8.18 8.06	21.2 20.8 20.5	1.35 1.30 1.26	3.43 3.30 3.20	111 94.5 81.6	1,820 1,550 1,340	9.35 7.64 6.37	153 125 104
W21 533 x 210 (21 x 8¼)	2,070 1,830.7 1,830 1,600 1,480 1,330 1,140.7 959.0	86,200 76,207 76,200 66,600 61,600 55,400 47,490 39,917	92.9 81.5 81.4 70.6 64.7 57.5 44.0 38.7	3,870 3,393 3,390 2,940 2,690 2,390 1,832 1,611	8.70 8.71 8.67 8.64 8.60 8.54 8.40 8.23	22.1 22.1 22.0 21.9 21.8 21.7 21.3 20.9	1.84 1.84 1.83 1.81 1.80 1.77 1.65 1.65	4.67 4.67 4.65 4.60 4.57 4.50 4.19 4.20	192 171 170.8 151 140 127 109.7 93.0	3,150 2,800 2,799 2,470 2,290 2,080 1,798 1,524	22.1 19.5 19.6 17.0 15.7 14.0 10.7 9.5	362 320.2 320 279 257 229 175.4 156
W21 533 x 311 (21 x 12¼)	3,630 3,220 2,960 2,670 2,420	151,000 134,000 123,000 111,000 101,000	376 333 305 274 248	15,600 13,900 12,700 11,400 10,300	9.17 9.12 9.09 9.05 9.02	23.3 23.2 23.1 23.0 22.9	2.95 2.93 2.92 2.90 2.89	7.49 7.44 7.42 7.37 7.34	329 295 273 249 227	5,390 4,830 4,470 4,080 3,720	60.1 53.5 49.2 44.5 40.3	985 877 806 729 660
W24 610 x 178 (24 x 7)	1,550 1,540 1,350	64,500 64,085 56,200	34.5 34.3 29.1	1,440 1,432 1,210	9.23 9.25 9.11	23.4 23.5 23.1	1.38 1.38 1.34	3.51 3.51 3.40	131.0 130.0 114.0	2,150 2,124 1,870	9.80 9.76 8.30	161 159.5 136
W24 610 x 229 (24 x 9)	1,830 2,100 2,370 2,700 4,418 3,806 3,397 3,003	76,200 87,400 98,600 112,000 183,900 158,400 141,400 125,000	70.4 82.5 94.4 109 181.2 154.1 136.2 119.4	2,930 3,430 3,930 4,540 7,544 6,415 5,670 4,972	9.55 9.69 9.79 9.87 10.12 10.08 1.00 9.96	24.3 24.6 24.9 25.1 25.7 25.6 25.4 25.3	1.87 1.92 1.95 1.98 2.05 2.03 2.00 1.98	4.75 4.88 4.95 5.03 5.21 5.15 5.09 5.04	154 176 196 222 349 304 274 245	2,520 2,880 3,210 3,640 5,718 4,990 4,495 4,012	15.7 18.4 20.9 24.0 39.3 33.8 30.1 26.5	157 302 342 393 644 554 493 435
W24 610 x 305 (24 x 12)	4,986.3 3,635.3 2,987.3	207,571 151,300 124,400	380.5 254.0 203.5	15,838 10,570 8,471	10.28 10.15 10.08	26.10 25.78 25.60	2.84 2.68 2.63	7.22 6.81 6.68	400.2 299.1 248.9	6,559 4,902.2 4,079.5	62.0 42.0 33.9	1,017 688.4 555.6
W24 610 x 324 (24 x 12¼)	6,256 5,660 5,170 4,580 4,020 3,540 3,100	260,400 236,600 215,000 191,000 167,000 147,000 129,000	529 479 443 391 340 297 259	22030 19950 18,400 16,300 14,200 12,400 10,800	10.6 10.5 10.4 10.3 10.2 10.1 10.1	26.8 26.6 26.4 26.2 25.9 25.7 25.7	3.07 3.04 3.05 3.01 2.97 2.94 2.91	7.79 7.73 7.75 7.65 7.54 7.47 7.39	491 450 414 371 329 291 258	8050 7380 6,780 6,080 5,390 4,770 4,230	81.8 74.3 68.4 60.5 53.0 46.5 40.7	1340 1218 1,120 991 869 762 667

Universal Beams And Columns



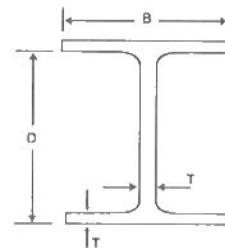
Section Number and Nominal Size	Unit weight M		Section area A		Section depth D		Flange				Web Thickness tw		Corner radius r	
							Width B		Thickness T					
mm (in)	lb/ft	kg/m	in ²	cm ²	in	mm	in	mm	in	mm	in	mm	in	mm
W27 686 x 254 (27 x 10)	182	271	53.5	345	28.50	723.9	10.244	260.2	1.539	39.1	0.850	21.6	0.60	15.2
	159	237	46.7	301	28.11	714.0	10.150	257.8	1.339	34.0	0.752	19.1	0.60	15.2
	143	213	41.8	270	27.87	707.9	10.071	255.8	1.220	31.0	0.669	17.0	0.60	15.2
	129	192	37.8	244	27.63	701.8	10.012	254.3	1.098	27.9	0.610	15.5	0.60	15.2
	114	170.0	33.5	216.1	27.29	693.2	10.070	255.8	0.930	23.62	0.570	14.48	0.60	15.2
	102	152.0	30.0	193.5	27.09	688.1	10.015	254.4	0.830	21.08	0.515	13.08	0.60	15.2
	94	140.0	27.7	178.7	26.92	683.8	9.990	253.7	0.745	18.92	0.490	12.45	0.60	15.2
	84	125.0	24.8	160.0	26.71	678.4	9.960	253.0	0.640	16.26	0.460	11.68	0.60	15.2
W27 686 x 356 (27 x 14)	217	322.9	63.7	411	28.43	722.1	14.114	358.5	1.500	38.1	0.831	21.1	0.60	15.2
	194	288.7	57.0	368	28.11	714.0	14.031	356.4	1.339	34.0	0.752	19.1	0.60	15.2
	178	264.9	52.3	337.4	27.81	706.4	14.085	357.8	1.190	30.23	0.725	18.41	0.60	15.2
	161	239.6	47.4	305.8	27.59	700.8	14.020	356.1	1.080	27.43	0.660	16.76	0.60	15.2
	146	217.3	42.9	276.8	27.38	695.5	13.965	354.7	0.975	24.76	0.605	15.37	0.60	15.2
W30 762 x 267 (30 x 10½)	185	275	54.4	351	31.22	793.0	10.634	270.1	1.461	37.1	0.811	20.6	0.65	16.5
	165	246	48.4	312	30.91	785.1	10.555	268.1	1.299	33.0	0.728	18.5	0.65	16.5
	148	220	43.6	281	30.67	779.0	10.480	266.2	1.181	30.0	0.650	16.5	0.65	16.5
	132	197.0	38.9	251.0	30.31	769.9	10.545	267.8	1.000	25.40	0.615	15.62	0.65	16.5
	124	185.0	36.5	235.5	30.17	766.3	10.515	267.1	0.930	23.62	0.585	14.86	0.65	16.5
	116	173.0	34.2	220.6	30.01	762.3	10.495	266.6	0.850	21.59	0.565	14.35	0.65	16.5
	108	161.0	31.7	204.5	29.83	757.7	10.475	266.1	0.760	19.30	0.545	13.84	0.65	16.5
	99	147.3	29.1	187.7	29.65	753.1	10.450	265.4	0.670	17.02	0.520	13.21	0.65	16.5
	90	133.9	26.5	171	29.53	750.0	10.409	264.4	0.610	15.5	0.472	12.0	0.65	16.5
	W30 762 x 381 (30 x 15)	261	388.4	76.7	495	31.61	802.9	15.154	384.9	1.650	41.9	0.929	23.6	0.65
235		349.7	69.0	445	31.30	795.0	15.055	382.4	1.500	38.1	0.831	21.1	0.65	16.5
211		314.0	62.0	400.0	30.94	785.9	15.105	383.7	1.310	33.40	0.775	19.68	0.65	16.5
191		284.2	56.1	361.9	30.68	779.3	15.040	382.0	1.180	30.10	0.710	18.03	0.65	16.5
173		257.5	50.8	327.7	30.44	773.2	14.985	380.6	1.060	27.05	0.655	16.64	0.65	16.5
W33 838 x 292 (33 x 11½)	204	304	59.8	386	34.30	871.2	11.642	295.7	1.461	37.1	0.811	20.6	0.70	17.8
	187	278	55.0	355	34.06	865.1	11.579	294.1	1.339	34.0	0.752	19.1	0.70	17.8
	169	252	49.4	319	33.82	859.0	11.500	292.1	1.220	31.0	0.669	17.0	0.70	17.8
	152	226.2	44.7	288.4	33.49	850.6	11.565	293.8	1.050	26.80	0.635	16.13	0.70	17.8
	141	210.0	41.6	268.4	33.30	845.8	11.535	293.0	0.960	24.38	0.605	15.37	0.70	17.8
	130	194.0	38.3	247.1	33.09	840.5	11.510	292.4	0.855	21.72	0.580	14.73	0.70	17.8
	118	176.0	34.7	223.9	32.86	834.6	11.480	291.6	0.740	18.80	0.550	13.97	0.70	17.8
W33 838 x 400 (33 x 15½)	291	433.1	85.6	552	34.84	884.9	15.906	404.0	1.728	43.9	0.961	24.4	0.70	17.8
	263	391.4	77.3	499	34.53	877.1	15.803	401.4	1.571	39.9	0.870	22.1	0.70	17.8
	241	358.6	70.9	457.4	34.18	868.2	15.860	402.8	1.400	35.56	0.830	21.08	0.70	17.8
	221	328.9	65.0	419.4	33.93	861.8	15.805	401.4	1.270	32.38	0.775	19.68	0.70	17.8
	201	299.1	59.1	381.3	33.68	855.5	15.745	399.9	1.150	29.21	0.715	18.16	0.70	17.8
W36 914 x 305 (36 x 12)	256	381	75.3	486	37.43	950.7	12.217	310.3	1.728	43.9	0.961	24.4	0.80	19.1
	232	345	68.2	440	37.12	942.9	12.118	307.8	1.571	39.9	0.870	22.1	0.80	19.1
	210	313.0	61.8	398.7	36.69	931.9	12.180	309.4	1.360	34.54	0.830	21.08	0.75	19.0
	194	289.0	57.0	367.7	36.49	926.8	12.115	307.7	1.260	32.00	0.765	19.43	0.75	19.0
	182	271.0	53.6	345.8	36.33	922.8	12.075	306.7	1.180	29.97	0.725	18.41	0.75	19.0
	170	253.0	50.0	322.6	36.17	918.7	12.030	305.6	1.100	27.94	0.680	17.27	0.75	19.0
	160	238.1	47.0	303.2	36.01	914.7	12.000	304.8	1.020	25.91	0.650	16.51	0.75	19.0
	150	224.0	44.2	285.2	35.85	910.6	11.975	304.2	0.940	23.88	0.625	15.87	0.75	19.0
	135	201.0	39.7	256.1	35.55	903.0	11.950	303.5	0.790	20.07	0.600	15.24	0.75	19.0
	W36 914 x 419 (36 x 16½)	359	534.2	105.4	680	37.40	950.0	16.728	424.9	2.012	51.1	1.122	28.5	0.95
328		488.1	96.4	622	37.10	942.1	16.630	422.4	1.850	47.0	1.020	25.9	0.95	24.1
300		447.0	88.3	569.7	36.74	933.2	16.655	423.0	1.680	42.67	0.945	24.00	0.95	24.1
280		417.0	82.4	531.6	36.52	927.6	16.595	421.5	1.570	39.88	0.885	22.48	0.95	24.1
260		388.0	76.5	493.5	36.26	921.0	16.550	420.4	1.440	36.58	0.840	21.34	0.95	24.1
245		365.0	72.1	465.2	36.08	916.4	16.510	419.4	1.350	34.29	0.800	20.32	0.95	24.1
230		343.0	67.6	436.1	35.90	911.9	16.470	418.3	1.260	32.00	0.760	19.30	0.95	24.1

Universal Beams And Columns (Continued)



Section Number and Nominal size	Moment of inertia				Radius of gyration				Modulus of section			
	I_x		I_y		r_x		r_y		Z_x		Z_y	
mm, (in)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
W27 686 x 254 (27 x 10)	6,950	289,300	277	11,540	11.4	29.0	2.28	5.78	488	7993	54.4	887
	5,953	247,800	234	9750	11.3	28.7	2.57	5.69	423	6940	46.1	756
	5,333	222,000	208	8678	11.3	28.7	2.23	5.67	383	6272	41.4	678
	4,757	198,000	184	7670	11.2	28.5	2.21	5.61	344	5644	36.8	603
	4,090	170,000	159	6,620	11.0	27.9	2.18	5.54	299	4,900	31.5	516
	3,620	151,000	139	5,790	11.0	27.9	2.15	5.46	267	4,380	27.8	456
	3,270	136,000	124	5,160	10.9	27.7	2.12	5.38	243	3,980	24.8	406
	2,850	119,000	106	4,410	10.7	27.2	2.07	5.26	213	3,490	21.2	347
W27 686 x 356 (27 x 14)	8,870	369,200	704	29,310	11.8	30.0	3.32	8.44	624	10,230	99.8	1635
	7,820	325,500	618	25,720	11.7	29.8	3.29	8.36	556	9,118	88.1	1,444
	6,990	291,000	555	23,100	11.6	29.5	3.26	8.28	502	8,230	78.8	1,290
	6,280	261,000	497	20,700	11.5	29.2	3.24	8.23	455	7,460	70.9	1,160
	5,630	234,000	443	18,400	11.4	29.0	3.21	8.15	411	6,740	63.5	1,040
W30 762 x 267 (30 x 10½)	8,486	353,200	294	12,240	12.5	31.7	2.33	5.91	544	8,909	55.3	906
	7,464	310,700	256	10,640	12.4	31.5	2.30	5.84	482	7,914	48.4	794
	6,684	278,200	227	9,463	12.4	31.5	2.29	5.81	436	7,142	43.4	711
	5,770	240,000	196	8,160	12.2	31.0	2.25	5.71	380	6,230	37.2	610
	5,360	223,000	161	7,530	12.1	30.7	2.23	5.66	355	5,820	34.4	564
	4,930	205,000	164	6,830	12.0	30.5	2.19	5.56	329	5,390	31.3	513
	4,470	186,000	146	6,080	11.9	30.2	2.15	5.46	299	4,900	27.9	457
	4,000	166,000	128	5,330	11.7	29.7	2.10	5.33	269	4,410	24.5	401
	3,621	150,700	115	4,788	11.7	29.7	2.09	5.30	245	4,018	22.1	362
W30 762 x 381 (30 x 15)	13,062	543,700	959	39,920	13.0	33.1	3.54	8.98	826	13,540	126.6	2,074
	11,674	485,900	855	35,570	13.0	33.0	3.52	8.94	746	12,220	113.5	1,860
	10,300	429,000	757	31,500	12.9	32.8	3.49	8.86	663	10,900	100	1,640
	9,170	382,000	670	28,000	12.8	32.5	3.46	8.79	598	9,800	89.5	1,470
	8,200	341,000	598	24,000	12.7	32.3	3.43	8.71	539	8,830	79.8	1,310
W33 838 x 292 (33 x 11½)	11,364	473,000	386	16,050	13.8	35.0	2.54	6.45	663	10,860	66.3	1,086
	10,340	430,400	346	14,470	13.7	34.8	2.51	6.38	607	9,950	60.0	984
	9,286	386,500	310	12,920	13.7	34.8	2.50	6.36	549	8,999	54.0	884
	8,160	340,000	273	11,400	13.5	34.3	2.47	6.27	487	7,980	47.2	773
	7,450	310,000	246	10,200	13.4	34.0	2.43	6.17	448	7,340	42.7	700
	6,710	279,000	218	9,070	13.2	33.5	2.38	6.05	406	6,640	37.9	621
	5,900	246,000	187	7,780	13.0	33.0	2.32	5.89	359	5,880	32.6	534
W33 838 x 400 (33 x 15¾)	17,668	735,400	1,163	48,390	14.4	36.5	3.69	9.36	1,104	16,620	146.2	2,396
	15,832	659,000	1,035	43,070	14.3	36.3	3.66	9.29	917	15,030	131.0	2,146
	14,200	591,000	932	38,800	14.1	35.8	3.63	9.22	829	13,600	116	1,930
	12,800	533,000	840	35,000	14.1	35.8	3.59	9.12	757	12,400	106	1,740
	11,500	479,000	749	31,200	14.0	35.6	3.56	9.04	684	11,200	95.2	1,560
W36 914 x 305 (36 x 12)	16,743	696,900	528	21,980	14.9	37.9	2.65	6.72	895	14,660	86.4	1,416
	15,020	625,200	468	19,480	14.8	37.7	2.62	6.66	809	13,260	77.3	1,266
	13,200	549,000	411	17,100	14.6	37.1	2.58	6.55	719	11,800	67.5	1,110
	12,100	504,000	375	15,600	14.6	37.1	2.56	6.50	664	10,900	61.9	1,010
	11,300	470,000	347	14,400	14.5	36.8	2.55	6.48	623	10,200	57.6	944
	10,500	437,000	320	13,300	14.5	36.8	2.53	6.43	580	9,500	53.2	872
	9,760	406,000	295	12,300	14.4	36.6	2.50	6.35	542	8,880	49.1	805
	9,040	376,000	270	11,200	14.3	36.3	2.47	6.27	504	8,260	45.1	739
	7,800	325,000	225	9,370	14.0	35.6	2.38	6.04	439	7,190	37.7	618
W36 914 x 419 (36 x 16½)	24,772	1,031,000	1,572	65,450	15.3	38.9	3.86	9.81	1,324	21,700	188	3,081
	22,497	936,400	1,422	59,170	15.3	38.8	3.84	9.76	1,213	19,880	171	2,801
	20,300	845,000	1,300	54,100	15.2	38.6	3.83	9.73	1,110	18,200	156	2,560
	18,900	787,900	1,200	49,900	15.1	38.4	3.81	9.68	1,030	16,900	144	2,360
	17,300	720,000	1,090	45,400	15.0	38.1	3.78	9.60	953	15,600	132	2,160
	16,100	670,000	1,010	42,000	15.0	38.1	3.75	9.52	895	14,700	123	2,020
	15,000	624,000	940	39,100	14.9	37.8	3.73	9.47	837	13,700	114	1,870

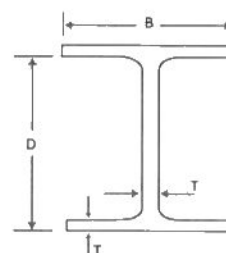
Universal Bearing Piles



IMPERIAL DIMENSIONS

US term	Serial size in	Weight per ft lb	Depth of Section D in	Width of Section B in	Thickness Web and Flange in	Area of section in ²
HP14	14 X 14½	117	14.234	14.885	0.805	34.40
		102	14.032	14.784	0.704	30.00
		89	13.856	14.696	0.616	26.10
		73	13.636	14.586	0.506	21.40
No imperial equivalent						
HP12	12 X 12	150	13.308	12.810	1.200	44.17
		125	12.928	12.620	1.010	36.73
		100	12.528	12.420	0.810	29.40
		85	12.308	12.310	0.700	24.60
		74	12.122	12.217	0.607	21.80
		64	11.960	12.136	0.526	18.40
		59	11.878	12.095	0.485	17.36
		53	11.780	12.046	0.436	15.50
No imperial equivalent						
HP10	10 X 10	57	10.012	10.224	0.564	16.80
		48	9.838	10.137	0.477	14.10
		42	9.72	10.078	0.418	12.40
No imperial equivalent						
HP8	8 X 8	36	8.026	81.58	0.446	10.60
		30	7.880	8.085	0.373	8.83
No imperial equivalent						

Universal Bearing Piles



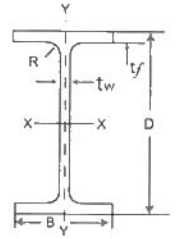
METRIC / IMPERIAL SIZES

Serial Size mm	Mass kg/m	Depth of Section D mm	Width of Section B mm	Thickness Web and Flange mm	Area of section cm ²
†400 X 400	235 197 168 140	408 400 394 388	412 408 405 402	25 21 18 15	300 251 214 178
*356 X 368	174 152 133 109	361.5 356.4 351.9 346.4	378.1 375.5 373.3 370.5	20.4 17.9 15.6 12.9	222.2 193.6 169.0 138.4
†350 X 350	156 131 106	350 344 338	357 354 351	19 16 13	198.4 166.6 135.3
*305 X 305	223 186 149 126 110 95 88 79	338.0 328.4 318.2 312.6 307.9 303.8 301.7 299.2	325.4 320.5 315.5 312.7 310.3 308.3 307.2 306.0	30.5 25.7 20.6 17.8 15.4 13.4 12.3 11.1	285.0 237.0 190.0 161.0 140.4 121.0 112.0 100.4
†300 X 300	142 106 84.5 83.4	310 300 294 294	310 305 302 302	20 15 12 12	181 134.8 107.7 106
*254 X 254	85 71 63	254.3 249.9 246.9	259.7 257.5 256.0	14.3 12.1 10.6	108.1 91.0 79.7
†250 X 250	82.2 64.4 63.8	250 244 244	255 252 252	14 11 11	104.7 82.06 81.3
*203 X 203	54 45	203.9 202.2	207.2 205.4	11.3 9.5	68.4 57.0
†200 X 200	56.2	200	204	12	71.53

*BS4: Part 1:1980 sizes

†JIS series

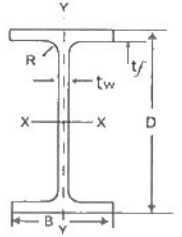
Universal Beams And Columns



Metric Units

Section size	Unit weight M	Section depth D	Flange width B	Tickness		Corner radius r	Section area A	Moment of inertia		Radius of gyration		Modulus of section	
				tw	tf			Ix	Iy	Ix	Iy	Zx	Zy
mm	kg/m	mm	mm	mm	mm	mm	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³
100 x 50	9.30	100	50	5	7	8	11.85	187	14.8	3.98	1.12	37.5	5.91
100 x 100	14.8	100	100	5	7	10	19.2	344	117.0	4.20	2.47	69.0	23.0
	16.9	100	100	8	6	8	21.6	378	134	4.18	2.49	75.6	26.7
	17.2	100	100	6	8	10	21.90	333	134	4.18	2.47	76.5	26.7
125 x 60	13.2	125	60	6	8	9	16.84	413	29.2	4.95	1.32	66.1	9.73
125 x 125	23.6	125	125	6.5	9	8	30.0	840	293	5.29	3.13	134	46.9
	23.8	125	125	6.5	9	10	30.31	847	293	5.29	3.11	136	47.0
150 x 75	14.0	150	75	5	7	8	17.85	666	49.5	6.11	1.66	88.8	13.2
150 x 100	20.7	148	100	6	9	8	26.3	1003	150	6.17	2.39	135	30.1
	21.1	148	100	6	9	11	26.84	1,020	151	6.17	2.37	138	30.1
150 x 150	31.1	150	150	7	10	8	39.6	1623	563	6.40	3.77	216	75.1
	31.5	150	150	7	10	11	40.14	1,640	563	6.39	3.75	219	75.1
	37.4	154	151	8	12	11	47.7	2018	690	6.51	3.80	262	91.4
175 x 90	18.0	175	90	5	8	8	22.9	1205	97.5	7.26	2.06	138	21.7
	18.1	175	90	5	8	9	23.04	1,210	97.5	7.26	2.06	139	21.7
175 x 125	23.3	169	125	5.5	8	12	29.65	1,530	261	7.18	2.97	181	41.8
175 x 175	32.8	171	174	6	9	12	41.7	2304	791	7.43	4.35	269	90.9
	40.2	175	175	7.5	11	13	51.2	2,880	984	7.50	4.38	330	112
	40.4	175	175	7.5	11	12	51.4	2895	984	7.50	4.37	331	112
200 x 100	17.8	198	99	4.5	7	8	22.7	1543	113	8.25	2.24	156	22.9
	18.2	198	99	7	4.5	11	23.2	1582	114	8.26	2.22	160	23.0
	20.9	200	100	5.5	8	8	26.7	1806	134	8.23	2.24	181	26.7
	21.3	200	100	5.5	8	11	27.16	1,840	134	8.24	2.22	184	26.8
200 x 150	29.9	194	150	6	9	8	38.1	2625	507	8.30	3.65	271	67.6
	30.6	194	150	6	9	13	39.01	2,690	507	8.30	3.61	277	67.6
	36.9	198	151	7	11	13	47.0	3331	633	8.42	3.67	336	83.8
200 x 200	41.4	196	199	6.5	10	13	52.7	3846	1315	8.54	5.00	392	132
	49.9	200	200	8	12	13	63.53	4,720	1,600	8.62	5.02	472	160
	56.2	200	204	12	12	13	71.53	4,980	1,700	8.35	4.88	498	167
	57.8	204	201	9	14	13	73.6	5603	1897	5.70	5.08	549	189
	65.7	208	202	16	10	13	83.69	6,530	2,200	8.83	5.13	628	218
250 x 125	25.1	248	124	5	8	8	32.0	3450	255	10.4	2.82	278	41.1
	25.7	248	124	5	8	12	32.68	3,540	255	10.4	2.79	285	41.1
	29.0	250	125	6	9	8	37.0	3965	294	10.4	2.82	317	47.0
	29.6	250	125	6	9	12	37.66	4,050	294	10.4	2.79	324	47.0
250 x 175	43.6	244	175	7	11	13	55.5	6037	984	10.4	4.21	495	112
	44.1	244	175	7	11	16	56.24	6120	985	10.4	4.19	502	113
	51.6	248	176	8	13	16	65.7	7308	1184	10.5	4.24	589	135
	59.1	252	177	9	15	16	75.3	8541	1390	10.7	4.30	678	157

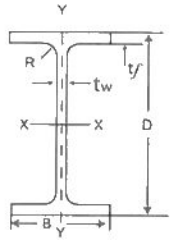
Universal Beams And Columns



Metric Units

Section size	Unit weight M	Section depth D	Flange width B	Tickness		Corner radius r	Section area A	Moment of inertia		Radius of gyration		Modulus of section	
				tw	tf			Ix	Iy	Ix	Iy	Zx	Zy
mm	kg/m	mm	mm	mm	mm	mm	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³
250 x 250	64.4	244	252	11	11	16	82.06	8,780	2,940	10.3	5.98	720	233
	66.5	248	249	8	13	16	84.7	9,930	3,350	10.8	6.29	801	269
	71.8	250	250	9	14	13	91.4	10750	3648	10.8	6.32	860	292
	72.4	250	250	9	14	16	92.18	10,800	3,650	10.8	6.29	867	292
	82.8	250	255	14	14	16	104.7	11,500	3,880	10.5	6.09	919	304
	98.1	260	253	12	19	16	125	15340	5134	11.1	6.41	1180	406
390 x 150	25.0	294	148	4.5	6	16	32.6	4940	326	12.3	3.16	336	44.0
	32.0	298	149	5.5	8	13	40.80	6,320	442	12.4	3.29	424	59.3
	36.7	300	150	6.5	9	13	46.78	7,210	508	12.4	3.29	481	67.7
	41.4	304	150	6.5	11	13	52.8	8578	620	12.7	3.43	564	82.7
	46.2	306	151	7.5	12	13	58.8	9514	691	12.7	3.43	622	91.5
	69.0	318	154	11	18	13	87.9	14820	1100	13.0	3.54	932	143
300 x 200	48.3	290	199	7	10	18	61.5	9431	1317	12.4	4.63	650	132
	55.8	294	200	8	12	13	71.1	11110	1602	12.5	4.75	756	160
	56.8	294	200	8	12	18	72.38	11,300	1,600	12.5	4.71	771	160
	65.4	298	201	9	14	18	83.36	13,300	1,900	12.6	4.77	893	189
	77.3	304	202	10	17	18	98.5	16280	2341	12.9	4.88	1071	232
300 x 300	84.5	294	302	12	12	18	107.7	16,900	5,520	12.5	7.16	1,150	365
	87.0	298	299	9	14	18	110.8	18,800	6,750	13.1	7.51	1,270	450
	93.0	300	300	10	15	13	118	20190	6753	13.1	7.55	1346	450
	94.0	300	300	10	15	18	119.8	20,400	6,750	13.1	7.51	1,360	450
	105	300	305	15	15	13	133.5	21,311	7,102	12.6	7.26	1,421	466
	106	304	301	11	17	18	134.8	23,400	7,730	13.2	7.57	1,540	514
	125	310	303	20	13	18	159	28130	9282	13.3	7.64	1815	613
	130	310	305	15	20	18	165	28630	9470	13.2	7.57	1847	621
	147	312	310	20	21	18	187	31370	10450	13.0	7.48	2011	674
350 x 175	41.2	346	174	6	9	13	52.5	11040	792	14.5	3.88	638	91.0
	41.4	346	174	6	9	14	52.68	11,100	798	14.5	3.88	641	91.0
	49.4	350	175	7	11	13	62.9	13500	984	14.6	3.96	771	113
	49.6	350	175	7	11	14	63.14	13,600	984	14.7	3.95	775	112
	57.8	354	176	8	13	14	73.7	16100	1184	14.8	4.01	909	135
	66.2	358	177	9	15	14	84.3	18710	1390	14.9	4.06	1045	157
	71.8	360	178	10	16	14	91.4	20240	1508	14.9	4.06	1124	169
	79.7	364	177	11	18	14	101	22760	1669	15.0	4.06	1250	189
350 x 250	69.2	336	249	8	12	20	88.15	18,500	3,090	14.5	5.92	1,100	248
	78.1	340	250	9	14	13	100	21230	3649	14.6	6.05	1249	292
	79.7	340	240	9	14	20	102	21,700	3,650	14.6	6.00	1,280	292
	94.2	346	251	10	17	20	120	26440	4488	14.8	6.12	1528	358
	108	350	253	12	19	20	137	30190	5138	14.8	6.12	1725	406

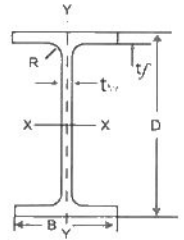
Universal Beams And Columns



Metric Units

Section size	Unit weight M	Section depth D	Flange width B	Tickness		Corner radius r	Section area A	Moment of inertia		Radius of gyration		Modulus of section	
				tw	tf			Ix	Iy	Ix	Iy	Zx	Zy
mm	kg/m	mm	mm	mm	mm	mm	mm	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³
350 x 350	106	338	351	13	13	20	135.3	28,200	9,380	14.4	8.33	1,670	534
	113	344	348	10	16	13	144	32,850	11,240	15.1	8.84	1,910	646
	115	344	348	10	16	20	146.0	33,810	11,840	15.1	8.95	1,966	646
	131	344	354	16	16	20	166.6	35,300	11,800	14.6	8.43	2,050	669
	135	350	350	12	19	13	172	39,847	11,840	15.2	8.89	2,277	776
	137	350	350	12	19	20	173.9	40,300	13,583	15.2	8.84	2,303	776
	156	350	357	19	19	20	198.4	42,500	14,400	14.7	8.53	2,450	809
	159	356	352	14	22	20	202.0	47,600	16,000	15.3	8.90	2,674	909
400 x 200	181	362	354	16	25	20	230	55,190	18,500	15.5	8.96	3,049	1,045
	56.1	396	199	7	11	13	71.4	19,770	1,447	16.6	4.50	999	145
	56.5	396	199	7	11	16	72.16	20,020	1,450	16.7	4.48	1,011	145
	65.4	400	200	13	8	13	83.4	23,460	1,736	16.8	4.56	1,173	174
	66.0	400	200	13	8	16	84.12	23,710	1,740	16.8	4.54	1,185	174
	75.5	404	201	9	15	16	96.2	27,490	2,035	16.9	4.60	1,361	202
	88.2	410	202	10	18	16	112	33,060	2,478	17.2	4.70	1,612	245
	140	430	208	16	28	16	179	54,850	4,216	17.5	4.86	2,551	405
400 x 300	94.3	386	299	9	14	22	120.1	33,700	6,240	16.7	7.21	1,740	418
	105	390	300	10	16	13	133	37,860	7,204	16.9	7.35	1,942	480
	107	390	300	10	16	22	136.0	38,680	7,210	16.9	7.28	1,983	481
	127	896	302	12	19	22	162	46,660	8,735	17.0	7.35	2,357	578
400 x 400	140	388	402	15	15	22	178.5	48,930	16,260	16.6	9.55	2,524	809
	147	394	398	11	18	22	186.8	56,150	18,930	17.3	10.1	2,850	951
	168	394	405	18	18	22	214.4	49,700	20,000	16.7	9.65	3,030	985
	172	400	400	13	21	22	218.7	66,620	22,420	17.5	10.1	3,331	1,120
	197	400	408	21	21	22	250.7	70,900	23,800	16.8	9.75	3,540	1,170
	200	406	403	16	24	22	254.9	78,040	26,200	17.5	10.1	3,844	1,300
	232	414	405	18	28	22	295.5	92,770	31,030	17.7	10.2	4,482	1,532
	283	428	407	20	35	22	360.7	119,000	39,360	18.2	10.4	5,570	1,934
	415	458	417	30	50	22	528.6	187,100	60,530	18.8	10.7	8,172	2,903
	605	598	432	45	70	22	770.1	297,900	94,360	19.7	11.1	12,000	4,370
450 x 200	65.1	446	199	8	12	13	83.0	28,130	1,579	18.4	4.36	1,262	159
	66.2	446	199	8	12	18	84.30	28,700	1,580	18.5	4.33	1,287	159
	74.9	450	200	9	14	13	95.4	32,890	1,870	18.6	4.43	1,462	187
	76.0	450	200	9	14	18	96.76	33,500	1,872	18.6	4.40	1,487	187
	88.9	456	201	10	17	18	113	40,400	2,308	18.9	4.51	1,772	230
	98.9	460	202	11	19	18	126	45,430	2,619	19.0	4.56	1,975	259
	110	460	205	14	19	18	140	47,860	2,742	18.5	4.43	2,081	268
	106	434	299	10	15	24	135.0	46,800	6,690	18.6	7.04	2,160	448
450 x 300	121	440	300	11	18	13	154	54,730	8,106	18.9	7.26	2,488	540
	124	440	300	11	18	24	157.4	56,100	8,110	18.9	7.18	2,550	541
	145	446	302	13	21	24	184	66,380	9,658	19.0	7.24	2,977	640

Universal Beams And Columns



Metric Units

Section size	Unit weight M	Section depth D	Flange width B	Tickness		Corner radius r	Section area A	Moment of inertia		Radius of gyration		Modulus of section	
				tw	tf			Ix	Iy	Ix	Iy	Zx	Zy
mm	kg/m	mm	mm	mm	mm	mm	mm	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³
500 x 200	77.9	496	199	9	14	13	99	40840	1843	20.3	4.31	1647	185
	79.5	496	199	9	14	20	101.3	41,900	1,846	20.3	4.27	1,690	186
	88.2	500	200	10	16	13	112	46810	2138	20.4	4.36	1872	214
	89.6	500	200	10	16	20	114.2	47,800	2,142	20.5	4.33	1,917	214
	102	506	201	11	19	13	129	55480	2578	20.7	4.47	2193	257
	103	506	201	11	19	20	131.3	56,520	2,582	20.7	4.43	2,234	257
	117	512	202	12	22	20	148	65450	3035	21.0	4.52	2557	300
500 x 300	111	482	300	11	15	13	141	58270	6756	20.3	6.92	2418	450
	114	482	300	11	15	26	145.5	60,400	6,768	20.4	6.82	2,505	451
	125	488	300	11	18	13	159	68860	8106	20.8	7.14	2822	540
	128	488	300	11	18	26	163.5	71,000	8,118	20.8	7.05	2,910	541
	150	494	302	13	21	26	191	83810	9663	20.9	7.11	3393	640
600 x 200	79.0	592	197	8	13	22	101	58240	1665	24.1	4.07	1968	169
	92.6	596	199	10	15	13	118	66640	1976	23.8	4.10	2236	199
	94.6	596	199	10	15	22	120.5	68,720	1,982	23.9	4.05	2,310	199
	103	600	200	11	17	13	132	75560	2274	24.0	4.16	2519	227
	106	600	200	11	17	22	134.4	77,630	2,280	24.0	4.12	2,588	228
	118	606	201	12	20	13	150	88320	2716	24.3	4.26	2915	270
	120	606	201	12	20	22	152.5	90,400	2,723	24.3	4.23	2,983	271
	134	612	202	13	23	22	171.7	103,500	3,180	24.6	4.32	3,382	315
600 x 300	133	582	300	12	17	13	169	98950	7659	24.2	6.73	3400	511
	137	582	300	12	17	28	174.5	102,700	7,675	24.3	6.63	3,530	512
	147	588	300	12	20	13	187	114400	9009	24.7	6.94	3889	601
	151	588	300	12	20	28	192.5	118,100	9,025	24.8	6.85	4,020	602
	170	594	302	14	23	13	217	133,650	10570	24.8	6.98	4497	700
	175	594	302	14	23	28	222.4	137,300	10,590	24.9	6.90	4,624	701
	203	602	304	16	27	28	259	162600	12680	25.1	7.00	5401	834
	217	608	304	16	30	28	277	179300	14090	25.4	7.13	5896	927
700 x 300	166	692	300	13	20	28	211.5	172,400	9,030	28.6	6.53	4,980	602
	182	700	300	13	24	18	232	197500	10820	29.2	6.83	5643	721
	185	700	300	13	24	28	235.5	201,500	10,830	29.3	6.78	5,760	722
	215	708	302	15	28	28	273.6	237,100	12,900	29.5	6.86	6,700	854
800 x 300	191	792	300	14	22	28	243.4	254,000	9,936	32.3	6.39	6,410	662
	207	800	300	14	26	18	264	286,400	11720	33.0	6.67	7159	781
	210	800	300	14	26	28	267.4	291,700	11,740	33.0	6.62	7,292	782
	241	808	302	16	30	28	307.6	339,200	13,820	33.2	6.70	8,400	915
	267	816	303	17	34	28	340	383600	15820	33.6	6.82	9402	1044
900 x 300	210	890	299	15	23	18	267	338500	10280	35.6	6.20	7608	687
	213	890	299	15	23	28	270.9	345,300	10,290	35.7	6.16	7,760	688
	240	900	300	16	28	18	306	404500	12630	36.4	6.43	8989	842
	243	900	300	16	28	28	309.8	411,300	12,650	36.4	6.39	9,140	843
	283	912	302	18	34	18	360	491000	15660	36.9	6.59	10770	1037
	286	912	302	18	34	28	364.0	497,800	15,670	37.0	6.56	10,920	1,038
	304	918	303	19	37	18	387	535400	17210	37.2	6.67	11660	1136
	307	918	303	19	37	28	391	542200	17230	37.2	6.63	11810	1137